

VIRASORO COHOMOLOGY AND HAMILTONIAN REDUCTION

1.1 THE STATE SPACE AND THE FIRST ANOMALY

A degree-one operator can define cohomology only when its degree-two square vanishes on the specified state space. For the Virasoro algebra, this square is a quadratic ghost operator. Its coefficient comes from a finite contraction in the ghost representation. The zero modes then determine which restrictions form cochain complexes.

All vector spaces and tensor products below are algebraic over $\mathbb{C}$. For homogeneous operators, the bracket is $[A, B]_s = AB - (-1)^{|A||B|}BA$. Braces denote the bracket of two odd operators. Matter operators have ghost degree zero.

Let $\mathcal{M}$ be a Virasoro module with central charge $C \in \mathbb{C}$:

$$[L_m^M, L_n^M] = (m - n)L_{m+n}^M + \frac{C}{12}(m^3 - m)\delta_{m+n,0} \mathbf{1}. \quad (1.1)$$

Assume that $\mathcal{M}$ is restricted: for each $v \in \mathcal{M}$, $L_n^M v = 0$ for all sufficiently large positive n . No inner product or finite-dimensional weight-space hypothesis is imposed.

The ghost Clifford algebra has odd generators $b_n, c_n, n \in \mathbb{Z}$, and relations

$$\{b_m, c_n\} = \delta_{m+n,0}, \quad \{b_m, b_n\} = \{c_m, c_n\} = 0.$$

Its conformal vacuum Ω obeys

$$b_n \Omega = 0 \quad (n \geq -1), \quad c_n \Omega = 0 \quad (n \geq 2). \quad (1.2)$$

The ghost module is

$$\mathcal{F} = \Lambda(b_n \ (n \leq -2), c_n \ (n \leq 1))\Omega.$$

A displayed generator acts by exterior multiplication. Its Clifford partner acts by left exterior differentiation. Thus the formulas specify the signs on every monomial. Set $\deg \Omega = 0$, $\deg c_n = 1$, and $\deg b_n = -1$. Set $E\Omega = 0$ and assign both b_n and c_n energy $-n$. The energy of a monomial is the sum of these assignments.

Normal ordering, denoted by colons, moves the creators in (1.2) to the left. Each interchange of odd factors contributes a minus sign. Contraction terms are omitted. This convention applies to the whole monomial inside the colons. Define

$$L_m^{\text{gh}} = \sum_{s \in \mathbb{Z}} (m + s) : b_{m-s} c_s :, \quad (1.3)$$

$$R = -\frac{1}{2} \sum_{m,n \in \mathbb{Z}} (m - n) : c_{-m} c_{-n} b_{m+n} :, \quad (1.4)$$

$$Q = \sum_{n \in \mathbb{Z}} c_{-n} L_n^M + R \quad \text{on } \mathcal{D} = \mathcal{M} \otimes \mathcal{F}. \quad (1.5)$$

Tensor factors commute because the matter action is even. The degree of Q is one.

LEMMA 1.1.1 (*Algebraic domain*). Every sum in (1.3)–(1.5) has finitely many nonzero terms on each element of $\mathcal{D}$. The same holds for

$$\nu_r = \frac{1}{2} \sum_{a+b=r} (a-b)c_a c_b, \quad K = \frac{C-26}{12} \sum_{n>0} (n^3-n)c_{-n}c_n.$$

These operators and all their finite compositions preserve $\mathcal{D}$.

Proof. It suffices to use one matter vector and one ghost monomial. For the matter term, indices $n \leq -2$ require the occupied creator b_n . Only finitely many such indices occur. The restrictedness of $\mathcal{M}$ bounds the remaining positive indices.

For a normally ordered ghost monomial, every annihilator must remove an occupied creator. Its indices therefore lie in a fixed finite set. All creator indices have an upper bound: -2 for b and 1 for c . The sum of the indices is fixed in each displayed operator. Once the annihilator indices are fixed, this sum and the upper bounds give lower bounds for every creator index. There are at most three factors. Hence only finitely many terms act nontrivially. The argument includes terms containing only creators. It also applies to ν_r and K . Each output is a finite linear combination of algebraic states, so compositions are defined on the same domain. $\square$

For trivial matter, all $L_n^{\mathcal{M}}$ vanish and $C = 0$. The first nonzero anomalous mode is visible without an infinite summation:

$$\begin{aligned} R\Omega &= 0, & Rb_{-2}\Omega &= -2b_{-2}c_0\Omega - b_{-3}c_1\Omega, \\ R(b_{-2}c_0\Omega) &= 4c_{-2}\Omega + b_{-3}c_0c_1\Omega + 2b_{-2}c_{-1}c_1\Omega, \\ R(b_{-3}c_1\Omega) &= 5c_{-2}\Omega - 2b_{-3}c_0c_1\Omega - 4b_{-2}c_{-1}c_1\Omega. \end{aligned}$$

Consequently,

$$R^2b_{-2}\Omega = -13c_{-2}\Omega. \quad (1.6)$$

The cancellation of the cubic output leaves a degree-two operator acting on a degree-minus-one input. The calculation below determines every mode of this operator.

1.2 THE GHOST CENTRAL TERM

LEMMA 1.2.1 (*Ghost commutants*). An operator on $\mathcal{M} \otimes \mathcal{F}$ that supercommutes with every b_n, c_n is zero if its ghost degree is nonzero. An even operator on $\mathcal{F}$ commuting with every b_n, c_n is a scalar.

Proof. The common kernel of the vacuum annihilators on $\mathcal{M} \otimes \mathcal{F}$ is $\mathcal{M} \otimes \mathbb{C}\Omega$. Indeed, these annihilators are all the exterior derivatives of the creator variables. A finite exterior polynomial killed by all these derivatives is constant. A supercommuting operator sends this common kernel into itself. The kernel has ghost degree zero, so an operator of nonzero ghost degree vanishes there. Supercommutation with the creators makes it vanish on every state. For an even operator on $\mathcal{F}$, its value on Ω is a scalar multiple of Ω . Commutation with the creators gives the same scalar on all monomials. $\square$

PROPOSITION 1.2.2 (*The ghost Virasoro algebra*). On $\mathcal{F}$,

$$[L_m^{\text{gh}}, b_n] = (m - n)b_{m+n}, \quad [L_m^{\text{gh}}, c_n] = (-2m - n)c_{m+n}, \quad (1.7)$$

$$L_0^{\text{gh}} = E, \quad L_m^{\text{gh}}\Omega = 0 \quad (m \geq -1), \quad (1.8)$$

$$[L_m^{\text{gh}}, L_n^{\text{gh}}] = (m - n)L_{m+n}^{\text{gh}} - \frac{26}{12}(m^3 - m)\delta_{m+n,0} \mathbf{1}. \quad (1.9)$$

Proof. The elementary contractions are

$$[: b_r c_s :, b_n] = \delta_{s+n,0} b_r, \quad [: b_r c_s :, c_n] = -\delta_{r+n,0} c_s.$$

They give (1.7). A term acting on Ω needs both factors to be creators. For $m \geq 0$ this is impossible. For $m = -1$ the only possible term has coefficient zero. The commutators of L_0^{gh} with the creators equal those of E . Both operators kill Ω , which proves (1.8).

Let $A_{m,n} = [L_m^{\text{gh}}, L_n^{\text{gh}}] - (m - n)L_{m+n}^{\text{gh}}$. Substitution of (1.7) into the Jacobi identity shows that $A_{m,n}$ commutes with every ghost generator. For example, its commutator with b_r has coefficient

$$(n - r)(m - n - r) - (m - r)(n - m - r) - (m - n)(m + n - r) = 0.$$

For c_r , the coefficient is

$$(-2n - r)(-2m - n - r) - (-2m - r)(-2n - m - r) - (m - n)(-2m - 2n - r) = 0.$$

Lemma 1.2.1 makes $A_{m,n}$ a scalar. Its energy is $-m - n$, so this scalar vanishes unless $m + n = 0$.

For $m \geq 2$, the complete vacuum expression is

$$L_{-m}^{\text{gh}}\Omega = \sum_{s=2-m}^{-1} (s - m)b_{-m-s}c_s\Omega.$$

Commuting L_m^{gh} through the first factor gives $(2m + s)b_{-s}$. This contracts with c_s to give 1. Its commutator with the second factor annihilates Ω . Therefore

$$\begin{aligned} A_{m,-m}\Omega &= \sum_{s=2-m}^{-1} (s - m)(2m + s)\Omega \\ &= \left(\sum_{s=2-m}^{-1} s^2 + m \sum_{s=2-m}^{-1} s - 2m^2 \cdot m \right) \Omega \\ &= -\frac{13}{6}(m^3 - m)\Omega. \end{aligned}$$

Here the interval has m integers, $\sum s = m(3 - m)/2$, and $\sum s^2 = m(2m^2 - 9m + 13)/6$. The cases $m = 0, 1$ give zero by (1.8). Antisymmetry gives negative m . This proves (1.9). $\square$

Set $\mathcal{L}_n = L_n^M + L_n^{\text{gh}}$ on $\mathcal{D}$. The two summands commute. Thus (1.1) and (1.9) give

$$[\mathcal{L}_m, \mathcal{L}_n] = (m - n)\mathcal{L}_{m+n} + A(m)\delta_{m+n,0}, \quad A(m) = \frac{C - 26}{12}(m^3 - m). \quad (1.10)$$

This central term determines the square of Q on the full ghost module.

1.3 THE SQUARE AS AN OPERATOR IDENTITY

LEMMA 1.3.1 (*Contractions of the charge*). The operator (1.5) satisfies

$$\{Q, b_n\} = \mathcal{L}_n, \quad \{Q, c_n\} = \nu_n. \quad (1.11)$$

Moreover,

$$[\mathcal{L}_m, \nu_n] = (-2m - n)\nu_{m+n}, \quad [Q, \nu_n] = 0. \quad (1.12)$$

Proof. Contracting b_n with the two c factors in R gives

$$\{R, b_n\} = \sum_{m \in \mathbb{Z}} (m - n) : c_{-m} b_{m+n} := \sum_{s \in \mathbb{Z}} (n + s) : b_{n-s} c_s := L_n^{\text{gh}}.$$

Contracting c_n with the b factor gives

$$\{R, c_n\} = -\frac{1}{2} \sum_{r+s=-n} (r - s) c_{-r} c_{-s} = \nu_n.$$

The matter term contributes L_n^M to the first identity and zero to the second. All contractions retain the same normal ordering of the uncontracted factors.

The commutator of $\mathcal{L}_m$ with ν_n uses (1.7). After reindexing, the coefficient of $c_u c_v$, $u + v = m + n$, is

$$\frac{1}{2} ((u - v - m)(-m - u) + (u - v + m)(-m - v)) = -\frac{1}{2} (2m + n)(u - v).$$

This proves the first part of (1.12).

For the second part, extend $\delta(c_n) = \nu_n$ as an odd derivation on products of c modes. Their mutual anticommutators are zero. The coefficient of each alternating cubic monomial in $\delta^2(c_n)$ is a multiple of

$$(a - b)(a + b - d) + (b - d)(b + d - a) + (d - a)(d + a - b) = 0.$$

Thus $\delta^2(c_n) = 0$. The second identity in (1.11) identifies $[Q, \nu_n]$ with this expression. Lemma 1.1.1 permits these expansions on each algebraic input. $\square$

THEOREM 1.3.2 (*The Virasoro square*). For every restricted matter module $\mathcal{M}$ as above,

$$Q^2 = \frac{C - 26}{12} \sum_{n > 0} (n^3 - n) c_{-n} c_n \quad \text{on } \mathcal{M} \otimes \mathcal{F}. \quad (1.13)$$

In addition,

$$[Q, \mathcal{L}_n] = -A(n)c_n, \quad [Q, \mathcal{L}_0] = 0. \quad (1.14)$$

If $\mathcal{M} \neq 0$, the operator Q is square-zero on the full state space if and only if $C = 26$.

Proof. Put $D_m = [Q, \mathcal{L}_m]$. The graded Jacobi identity, (1.11), and (1.10) give

$$\begin{aligned} \{D_m, b_n\} &= [Q, [\mathcal{L}_m, b_n]]_s - [\mathcal{L}_m, \{Q, b_n\}] \\ &= (m - n)\mathcal{L}_{m+n} - [\mathcal{L}_m, \mathcal{L}_n] = -A(m)\delta_{m+n,0}, \\ \{D_m, c_n\} &= (-2m - n)\nu_{m+n} - [\mathcal{L}_m, \nu_n] = 0. \end{aligned}$$

Hence $D_m + A(m)c_m$ supercommutes with all ghosts. It has ghost degree one, so Lemma 1.2.1 makes it zero. This proves (1.14).

Let K be the right side of (1.13). A direct Clifford contraction gives

$$[K, b_n] = -A(n)c_n, \quad [K, c_n] = 0.$$

For instance, $[c_{-m}c_m, b_n] = \delta_{m+n,0}c_{-m} - \delta_{n-m,0}c_m$. The oddness of A gives the formula for both signs of n . Also,

$$[Q^2, b_n] = [Q, \{Q, b_n\}] = [Q, \mathcal{L}_n], \quad [Q^2, c_n] = [Q, v_n] = 0.$$

Thus $Q^2 - K$ commutes with all ghosts and has ghost degree two. Lemma 1.2.1 proves that it is zero.

When $C = 26$, the displayed square vanishes. For nonzero $v \in M$, the converse follows from

$$Q^2(v \otimes b_{-2}\Omega) = \frac{C-26}{2} v \otimes c_{-2}\Omega.$$

The output is nonzero when $C \neq 26$. □

1.4 FIELDS, ZERO MODES, AND THE INTERCEPT

Fix a formal coordinate z and define

$$b(z) = \sum_n b_n z^{-n-2}, \quad c(z) = \sum_n c_n z^{1-n}, \quad T^M(z) = \sum_n L_n^M z^{-n-2}.$$

The residue extracts the coefficient of $z^{-1}dz$. For convergent Laurent series it equals the positively oriented integral divided by $2\pi i$. All uses here are formal. The two ghost contractions are the expansion of $(z-w)^{-1}$ in nonnegative powers of w/z . For example,

$$\sum_{n \geq -1} z^{-n-2} w^{n+1} = \frac{1}{z-w}.$$

The ghost stress tensor is

$$T^{\text{gh}}(z) = -2 : b(z) \partial c(z) : - : \partial b(z) c(z) : = \sum_m L_m^{\text{gh}} z^{-m-2}.$$

Indeed, the coefficient of $b_r c_s$ is $r + 2s$. When $r + s = m$, this is $m + s$, as in (1.3). Equation (1.7) specifies the conformal weights 2 and -1.

Multiplying the fields and taking the residue gives exactly

$$Q = \text{Res}_{z=0} (c(z) T^M(z) + : b(z) c(z) \partial c(z) :) dz. \quad (1.15)$$

For the matter term, the residue condition is $m + n = 0$. For the ghost term it is $r + s + t = 0$. Antisymmetrizing the two c factors changes the coefficient $1 - t$ to $(s - t)/2$, which gives (1.4). If $\mathcal{N}$ means full normal ordering of the elementary ghost factors, then

$$\frac{1}{2} \mathcal{N}(c T^{\text{gh}}) = : bc \partial c :.$$

Thus (1.15) also gives the residue of $c T^M + \frac{1}{2} \mathcal{N}(c T^{\text{gh}})$. Adding $\frac{3}{2} \partial^2 c$ to this current leaves its residue unchanged. This specifies a zero mode, rather than a sum of all current modes.

PROPOSITION I.4.1 (*Intercept and vacuum conventions*). For $t \in \mathbb{C}$, set $Q_t = Q + tc_0$. Then

$$\{Q_t, b_n\} = \mathcal{L}_n + t\delta_{n,0}, \quad [Q_t, \mathcal{L}_0] = 0, \quad (I.16)$$

$$Q_t^2 = \sum_{n>0} \left(\frac{C-26}{12} (n^3 - n) - 2tn \right) c_{-n}c_n. \quad (I.17)$$

For $M \neq 0$, absolute nilpotence holds exactly when $C = 26$ and $t = 0$.

Let $\downarrow = c_1\Omega$ and normal-order relative to this vacuum. Its annihilators are b_n for $n \geq 0$ and c_n for $n \geq 1$. Use a subscript $\downarrow$ for this ordering. Then

$$L_{n,\downarrow}^{\text{gh}} = L_n^{\text{gh}} + \delta_{n,0}, \quad R_{\downarrow} = R + c_0, \quad \sum_n c_{-n}L_n^M + R_{\downarrow} - ac_0 = Q + (1-a)c_0. \quad (I.18)$$

In this convention the nilpotent intercept is $a = 1$. The underlying Clifford module is the same. The vacuum $\downarrow$ has conformal energy -1 and ghost degree one in the Ω convention.

Proof. The first line follows from (I.11) and $[\mathcal{L}_0, c_0] = 0$. Since $c_0^2 = 0$,

$$Q_t^2 = Q^2 + t\{Q, c_0\}, \quad \{Q, c_0\} = \nu_0 = -2 \sum_{n>0} nc_{-n}c_n.$$

This proves (I.17). On $v \otimes \Omega$, its value is $-2t v \otimes c_{-1}c_1\Omega$. Thus absolute nilpotence requires $t = 0$. Theorem I.3.2 then requires $C = 26$.

The change of vacuum reverses only the creator/annihilator pair (c_1, b_{-1}) . Its normal products satisfy

$$: b_{-1}c_1 :_{\downarrow} - : b_{-1}c_1 := b_{-1}c_1 + c_1b_{-1} = 1.$$

The coefficient of this pair in L_0^{gh} is 1, and it occurs in no other L_n^{gh} . This gives the first identity in (I.18). The same contraction proof as in Lemma I.3.1 gives

$$\{R_{\downarrow} - R, b_n\} = \delta_{n,0}, \quad \{R_{\downarrow} - R, c_n\} = 0.$$

The difference $R_{\downarrow} - R - c_0$ vanishes by Lemma I.2.1. The last identity in (I.18) follows. The energy and degree of $\downarrow$ follow from those of c_1 . $\square$

PROPOSITION I.4.2 (*The nonnegative ghost polarization*). Set $c^m = c_{-m}$, so that $\{b_m, c^n\} = \delta_m^n$. Let $\mathcal{F}_{\text{pol}}$ have vacuum Ω_{pol} killed by c^m for $m \leq 0$ and by b_m for $m > 0$. Its creators are c^m for $m > 0$ and b_m for $m \leq 0$. Give its vacuum degree and energy zero, and assign these creators degrees 1, -1 and energies $m, -m$ respectively. There is a Clifford-module isomorphism

$$I : \mathcal{F}_{\text{pol}} \longrightarrow \mathcal{F}, \quad I(x\Omega_{\text{pol}}) = x c_0 c_1 \Omega,$$

where x is a product of the indicated creators. It raises ghost degree by 2 and lowers energy by 1. Let R_{pol} denote the cubic expression (I.4) with this ordering. Under $I_M \otimes I$,

$$\sum_m c^m L_m^M + R_{\text{pol}} - ac^0 \longmapsto Q + (1-a)c_0.$$

Thus its critical charge term is $-c^0$, with both degree and energy shifts specified by I .

Proof. The vector $\Xi = c_0 c_1 \Omega$ obeys the required annihilator conditions. For $m \leq 0$, $c_{-m} \Xi = 0$: c_0 and c_1 square to zero, and all c_n with $n \geq 2$ annihilate Ξ . For $m > 0$, $b_m \Xi = 0$ by the Clifford relations and (1.2). Hence the displayed map intertwines the Clifford actions. Away from the pairs (b_0, c_0) and (b_{-1}, c_1) it identifies the same exterior creators. On each of these two pairs, exterior multiplication and exterior differentiation exchange the empty and occupied states, with nonzero signs. Their four-state tensor product is invertible. This proves that I is an isomorphism. Since Ξ has degree 2 and energy -1 , the grading statements follow.

Relative to Ω , this ordering reverses both specified pairs. The pair (b_{-1}, c_1) contributes 1 to L_0^{gh} , as in Proposition 1.4.1. The pair (b_0, c_0) has coefficient zero there. Consequently the transported ghost stress modes are $L_n^{\text{gh}} + \delta_{n,0}$. The contraction identities and Lemma 1.2.1 then give $I R_{\text{pol}} I^{-1} = R + c_0$. The matter term commutes with I , which proves the charge formula. $\square$

1.5 ABSOLUTE AND RELATIVE COMPLEXES

Here absolute and relative refer to the b_0 constraint. The central character of the Virasoro module is fixed, and the ghost variables are indexed by the modes L_n . There is no ghost for the central generator in this carrier.

Set

$$\mathcal{D}_t^{\text{rel}} = \ker(b_0 : \mathcal{D} \rightarrow \mathcal{D}) \cap \ker(\mathcal{L}_0 + t : \mathcal{D} \rightarrow \mathcal{D}).$$

It inherits the ghost grading of $\mathcal{D}$. The first identity in (1.16) and the commutation with $\mathcal{L}_0$ show that Q_t preserves this subspace. On it, the square is the restriction of (1.17).

COROLLARY 1.5.1 (*The critical complexes*). At $C = 26$ and $t = 0$, the pairs

$$C_{\text{abs}}^q(M) = M \otimes \mathcal{F}^q, \quad C_{\text{rel}}^q(M) = \mathcal{D}_0^{\text{rel}} \cap (M \otimes \mathcal{F}^q)$$

with differential Q are cochain complexes of complex vector spaces. The inclusion $C_{\text{rel}}^\bullet(M) \rightarrow C_{\text{abs}}^\bullet(M)$ is a chain map. A homomorphism of restricted Virasoro modules induces chain maps by tensoring with $1_{\mathcal{F}}$. If $\mathcal{L}_0$ acts semisimply, every nonzero weight λ subcomplex of the absolute complex is contractible by $\lambda^{-1} b_0$.

Proof. The domain lemma defines the differential in every degree. Theorem 1.3.2 gives its square, and (1.16) gives stability of the relative subspace. A module homomorphism commutes termwise with (1.5), hence with the finite sums on each state. On the weight- λ subspace, $Q(\lambda^{-1} b_0) + (\lambda^{-1} b_0)Q = 1$ by (1.11). $\square$

Relative nilpotence depends on the actual relative carrier. The full-state converse in Theorem 1.3.2 cannot be restricted without checking its detecting vectors.

PROPOSITION 1.5.2 (*Nonnegative matter weights*). Suppose $M = \bigoplus_{j \geq 0} M_j$, with $L_0^M|_{M_j} = j$ and $L_n^M M_j \subseteq M_{j-n}$. At $t = 0$, for every central charge C , the relative carrier is

$$\mathcal{D}_0^{\text{rel}} = (M_0 \otimes \mathbb{C}\Omega) \oplus (M_1 \otimes \mathbb{C}c_1\Omega) \oplus (M_0 \otimes \mathbb{C}c_{-1}c_1\Omega). \quad (1.19)$$

Its differential is the three-term complex

$$M_0 \xrightarrow{L_{-1}^M} M_1 \xrightarrow{L_1^M} M_0 \quad (1.20)$$

in ghost degrees 0, 1, 2. In particular, $Q^2 = 0$ on this relative carrier for every C .

Proof. The only negative ghost energy is that of c_1 , and it can occur at most once. Thus $\mathcal{F}$ has minimum energy -1 . Every b creator has energy at least 2, so no b creator occurs at ghost energy 0 or -1 . The complete bases at these two energies are

$$\mathcal{F}_{-1} = \langle c_1 \Omega, c_0 c_1 \Omega \rangle, \quad \mathcal{F}_0 = \langle \Omega, c_0 \Omega, c_{-1} c_1 \Omega, c_{-1} c_0 c_1 \Omega \rangle.$$

The condition $b_0 = 0$ removes precisely the terms containing c_0 . The total weight-zero condition therefore gives (1.19).

The matter term and ghost term of (1.5) give

$$\begin{aligned} Q(v \otimes \Omega) &= L_{-1}^M v \otimes c_1 \Omega & (v \in M_0), \\ Q(w \otimes c_1 \Omega) &= L_1^M w \otimes c_{-1} c_1 \Omega & (w \in M_1), \\ Q(v \otimes c_{-1} c_1 \Omega) &= 0 & (v \in M_0). \end{aligned}$$

In the middle line, $c_0 L_0^M w \otimes c_1 \Omega$ cancels $w \otimes R c_1 \Omega = -w \otimes c_0 c_1 \Omega$. For the last line, $R(c_{-1} c_1 \Omega) = 0$ by the two contractions in Lemma 1.3.1. Finally, $L_1^M L_{-1}^M v = 2L_0^M v + L_{-1}^M L_1^M v = 0$ on M_0 . This proves (1.20) and its nilpotence. $\square$

For example, the universal Virasoro vacuum module of central charge C is induced from a vector $\mathbf{1}$ killed by L_n for $n \geq -1$. The central generator acts by C . The ordered monomials in L_{-n} , $n \geq 2$, form its basis by the Poincaré–Birkhoff–Witt theorem. Hence $M_0 = \mathbb{C}\mathbf{1}$ and $M_1 = 0$. Its relative complex has basis $\mathbf{1} \otimes \Omega$, $\mathbf{1} \otimes c_{-1} c_1 \Omega$ and zero differential for every C . Trivial matter gives the same relative complex at $C = 0$. Its absolute square is nonzero by (1.6). At the other boundary, suppose a restricted module contains $v \neq 0$ with $L_0^M v = -2v$. Then $v \otimes b_{-2} \Omega$ lies in $\mathcal{D}_0^{\text{rel}}$ and

$$Q^2(v \otimes b_{-2} \Omega) = \frac{C - 26}{2} v \otimes c_{-2} \Omega.$$

This relative square is nonzero when $C \neq 26$. Thus the critical central charge guarantees relative nilpotence for every restricted module, while particular relative carriers can have further vanishing.

1.6 CONFORMAL VERTEX ALGEBRA STATES

Let V be an ordinary conformal vertex algebra over $\mathbb{C}$ with conformal vector ω . Write $Y(\omega, z) = \sum_n L_n^V z^{-n-2}$ and let C be its central charge. The lower truncation axiom makes V a restricted matter module. Consequently (1.5) defines the degree-one operator on the full algebraic state carrier $V \otimes \mathcal{F}$. Equation (1.15) identifies it with the zero mode of the specified normal-ordered current. At $C = 26$, Corollary 1.5.1 supplies both the absolute complex and its b_0 -relative subcomplex. When V has nonnegative integral conformal weights, Proposition 1.5.2 computes the relative complex directly.

The ghost degree here is the cohomological degree. Conformal weight is the eigenvalue of $L_0^V + L_0^{\text{gh}}$ and is independent of ghost degree. The residue uses the fixed formal coordinate z . A comparison with a complex of collision-state sheaves requires maps in that sheaf category, with compatible restrictions and differentials. The state-space construction above supplies its Virasoro differential and its square.

1.6.1 Augmentations and the relative coordinate carrier

The critical Virasoro charge acts on an algebraic state space. A comparison with chiral chains also needs a target and compatible coordinate transitions. Two finite identities obstruct the augmentation-kernel target and the standard restricted transitions.

PROPOSITION 1.6.1 (*The augmentation-kernel obstruction*). Let $V \neq 0$ be a unital vertex algebra over $\mathbb{C}$, and let $A = V \otimes bc$ use the ghost system above. There is no unital vertex superalgebra homomorphism $A \rightarrow \mathbb{C}$ with trivial target vertex products. Consequently the reduced chiral construction whose coefficient sheaf is a chiral augmentation kernel cannot be instantiated on A with its original collision products.

Proof. Write $\Omega_A = \mathbf{1}_V \otimes \Omega$, $b = \mathbf{1}_V \otimes b_{-2}\Omega$, and $c = \mathbf{1}_V \otimes c_1\Omega$ in A . With the vertex convention $Y(a, z) = \sum_r a_{(r)} z^{-r-1}$,

$$b_{(0)}c = \mathbf{1}_V \otimes b_{-1}c_1\Omega = \Omega_A.$$

Every nonnegative product vanishes in the trivial vertex algebra. A unital homomorphism ε would therefore give $1 = \varepsilon(b_{(0)}c) = \varepsilon(b)_{(0)}\varepsilon(c) = 0$. For $V = \text{Vir}_{26}$, set $T = (L_{-2}^V \mathbf{1}_V) \otimes \Omega$. There is also the independent matter identity

$$T_{(3)}T = (L_2^V L_{-2}^V \mathbf{1}_V) \otimes \Omega = 13\Omega_A.$$

On a coordinate disk, let $t = z_1 - z_2$ and normalize $\text{Res}(t^{-1}dt) = 1$. The full collision maps include

$$\text{Res}_{t=0} Y(b, t)c dt = \Omega_A, \quad \text{Res}_{t=0} t^3 Y(T, t)T dt = 13\Omega_A.$$

A chiral augmentation to the unit chiral algebra must preserve these residues, which gives the same contradiction on that disk. It therefore cannot exist globally. The forms $dt = t d \log t$ and $t^3 dt = t^4 d \log t$ have regular logarithmic coefficients. Restricting to such coefficients does not remove the obstruction.

The linear vacuum projection $\pi : A \rightarrow \mathbb{C}$ has $b, c \in \ker \pi$ but $b_{(0)}c \notin \ker \pi$. Its kernel is not closed under the original collision maps. The quotient by the vacuum also fails to inherit all vertex products, since $(\Omega_A)_{(-1)}b = b$ whereas $0_{(-1)}b = 0$. Thus neither linear operation supplies the missing chiral augmentation. $\square$

PROPOSITION 1.6.2 (*Relative states, absolute states, and coordinate changes*). Take $V = \text{Vir}_{26}$ and the zero-intercept charge of Theorem 1.3.2. In this proposition, Ω denotes the tensor-product vacuum $\mathbf{1}_V \otimes \Omega_{\text{gh}}$. Put $H = L_o^V + L_o^{\text{gh}}$ and

$$e = \Omega, \quad u = c_o\Omega, \quad v = c_{-1}c_1\Omega, \quad w = c_{-1}c_o c_1\Omega.$$

The weight-zero absolute complex has these four basis states and differential

$$Qe = 0, \quad Qu = -2v, \quad Qv = Qw = 0.$$

The relative complex is $\mathbb{C}e \oplus \mathbb{C}v$ with zero differential. The absolute cohomology has one class in degrees zero and three, and no others. The inclusion of relative states into absolute states kills $[v]$ and is not a quasi-isomorphism.

Fix the convention that the coordinate flow $z \mapsto z/(1 + sz)$ acts by $g_s = \exp(s\mathcal{L}_1)$, where $\mathcal{L}_1 = L_1^V + L_1^{\text{gh}}$. Then

$$g_s v = v - s c_o c_1 \Omega, \quad b_o g_s v = -s c_1 \Omega \neq 0 \quad (s \neq 0). \quad (1.21)$$

The fixed relative fiber therefore does not define a subbundle through restriction of these state-bundle transitions.

Proof. The universal vacuum module has $V_0 = \mathbb{C}\mathbf{1}$ and $V_1 = 0$. The energy argument of Proposition 1.5.2 gives exactly the four displayed absolute states and the two relative states. The charge contraction gives $Qu = -2v$. Nilpotence gives $Qv = 0$, and weight zero has no degree-four state, so $Qw = 0$. At every nonzero total weight λ , the homotopy b_0/λ contracts the absolute complex by Corollary 1.5.1. On the four-state complex, the homotopy $h(v) = -u/2$, zero on e, u, w , contracts it onto $\mathbb{C}e \oplus \mathbb{C}w$. All state sums are algebraic, so these homotopies define maps on the full carrier. In particular, $v = Q(-u/2)$ in the absolute complex.

The ghost transformation law (1.7) gives

$$\mathcal{L}_1 v = -c_0 c_1 \Omega, \quad \mathcal{L}_1(c_0 c_1 \Omega) = 0.$$

The exponential on v consequently terminates after its linear term. Since $b_0 c_0 c_1 \Omega = c_1 \Omega$, this proves (1.21). Also $H g_5 v = s c_0 c_1 \Omega$, so the same transition fails the weight-zero condition. A restricted transition must preserve the model fiber, which fails here. $\square$

For a fixed coordinate, the relative vector-space complex remains defined. A sheaf comparison requires a separately constructed relative sheaf complex or a family of transported constraints with compatible transition maps. For source and target sheaf complexes $\mathcal{C}, \mathcal{T}$, local maps must satisfy

$$d_{\mathcal{T}} \Phi_U = \Phi_U d_{\mathcal{C}}, \quad \text{res}_{UW}^{\mathcal{T}} \Phi_U = \Phi_W \text{res}_{UW}^{\mathcal{C}} \quad (W \subset U),$$

and $g_{ij}^{\mathcal{T}} \Phi_j = \Phi_i g_{ij}^{\mathcal{C}}$ in coordinate trivializations. Ordinary or derived global sections require these complexes and their transitions first. The restricted transitions in (1.21) do not supply them.

A different unital chiral-chain construction may avoid an augmentation kernel. Its coefficient complexes, vacuum insertions, all collision maps, differential, and section functor must be constructed before it can serve as a comparison target. For the relative vacuum complex, a degree-preserving quasi-isomorphism requires target cohomology precisely in degrees zero and two. Proposition 1.6.2 excludes the absolute state complex as a replacement target. These assertions concern the specified augmentation-kernel construction and state transitions; they do not exclude every chiral-chain construction.

1.7 HAMILTONIAN REDUCTION AND COMPARISON OF COMPLEXES

The constraint $e = 1$ for $\mathfrak{sl}_2$ gives a reduction whose maps can be written on every polynomial. It separates Hamiltonian reduction from the Virasoro charge and from chiral collision operations. All complexes in the polynomial calculation are algebraic cochain complexes over $\mathbb{C}$. Their products use the Koszul sign $(-1)^{|x||y|}$.

1.7.1 The polynomial reduction

Let e, h, f be the linear coordinate functions on $\mathfrak{sl}_2^*$, with Poisson brackets

$$\{h, e\} = 2e, \quad \{h, f\} = -2f, \quad \{e, f\} = h.$$

Introduce odd generators b, c of cohomological degrees $-1, 1$. The even generators have degree zero. On the graded commutative algebra

$$A = \mathbb{C}[e, h, f] \otimes \Lambda(b, c)$$

define a degree-one derivation D by

$$De = 0, \quad Dh = -2ec, \quad Df = hc, \quad Db = e - 1, \quad Dc = 0.$$

Thus the Koszul differential imposes $e = 1$, and the other term is the Chevalley–Eilenberg differential for the Hamiltonian vector field $\{e, -\}$. The two terms anticommute. Direct substitution on the five generators gives $D^2 = 0$. Let $R = \mathbb{C}[u]$ have zero differential and augmentation $u \mapsto 0$. Give A the augmentation

$$\varepsilon(e) = 1, \quad \varepsilon(h) = \varepsilon(f) = \varepsilon(b) = \varepsilon(c) = 0.$$

These choices concern the displayed differential graded commutative algebra. They do not give an augmentation of an affine or ghost vertex algebra.

THEOREM 1.7.1 (*Polynomial reduction and its bar comparison*). The maps of augmented differential graded algebras

$$i : R \longrightarrow A, \quad i(u) = ef + \frac{1}{4}h^2, \quad (1.22)$$

$$p : A \longrightarrow R, \quad (p(e), p(h), p(f), p(b), p(c)) = (1, 0, u, 0, 0) \quad (1.23)$$

satisfy $pi = 1$ and are quasi-isomorphisms. In particular,

$$H^j(A, D) = \begin{cases} \mathbb{C}[u], & j = 0, \\ 0, & j \neq 0. \end{cases}$$

The induced maps $B(i)$ and $B(p)$ on the direct-sum associative reduced bar constructions are quasi-isomorphisms, and $B(p)B(i) = 1$.

Proof. The element $K = ef + h^2/4$ satisfies $DK = ehc - ehc = 0$. The projection kills the image under D of every generator, including $Db = e - 1$. This proves the chain identities and $pi = 1$.

Resolve the differential as

$$D = (e - 1)\partial_b + c(-2e\partial_h + h\partial_f).$$

Filter by the number of c factors. The filtration has two steps, so its spectral sequence has no convergence issue. Its first cohomology calculation is the two-term Koszul complex for $e - 1$. Multiplication by $e - 1$ is injective on the polynomial ring. The resulting complex is

$$\mathbb{C}[h, f] \xrightarrow{-2\partial_h + h\partial_f} c\mathbb{C}[h, f].$$

Set $u = f + h^2/4$. This is a polynomial change of coordinates from (h, f) to (h, u) . In these coordinates the displayed derivative is $-2\partial_h$. Its kernel is $\mathbb{C}[u]$, and it is surjective in characteristic zero. For example, the primitive of $ch^a u^j$ is $-h^{a+1}u^j/(2a+2)$. Thus cohomology is concentrated in degree zero, and p induces its displayed identification. The identity $pi = 1$ proves the same assertion for i .

Here are the bar conventions and the induced maps. For an augmented differential graded algebra E , put $\bar{E} = \ker \varepsilon$ and

$$B(E) = \bigoplus_{n \geq 0} (\bar{E}[1])^{\otimes n}.$$

Write sa for a considered in $\bar{E}[1]$, so $|sa| = |a| - 1$. The bar coderivation has components

$$b_1(sa) = -s(d_E a), \quad b_2(sa \otimes sb) = (-1)^{|a|} s(ab).$$

Extend each component to consecutive positions with the Koszul sign for a degree-one operator. The Leibniz rule and associativity give $b^2 = 0$. For an augmented chain algebra map q , set

$$B(q)(sa_1 \otimes \cdots \otimes sa_n) = sq(a_1) \otimes \cdots \otimes sq(a_n).$$

The identities $qd_E = d_F q$ and $q(ab) = q(a)q(b)$ prove that $B(q)$ commutes with b_1 and b_2 separately.

Filter the cone of $B(p)$ by tensor length. Its length- n quotient is the cone of $(\tilde{p}[1])^{\otimes n}$ with the internal differentials. The splitting $E = \mathbb{C} \oplus \tilde{E}$ shows that $\tilde{p}$ is a quasi-isomorphism. Tensor products of complexes of vector spaces preserve quasi-isomorphisms. One proof splits every acyclic complex into two-term contractible summands and tensors its contraction with the identity. Consequently each length quotient of the cone is acyclic. A cone cycle has finite maximal length. Solve its boundary equation in that highest quotient and subtract the resulting boundary. This lowers its maximal length. Induction ends at length zero and proves that the whole cone is acyclic. The same argument applies to $B(i)$, and functoriality gives $B(p)B(i) = 1$. $\square$

The bar argument uses finite tensor words and polynomial primitives. A product completion, a vertex algebra, and a chiral sheaf each require a separate comparison. Even on $\mathcal{A}$, the projection p does not preserve the original Poisson brackets:

$$p\{h, f\} = -2u, \quad \{p(h), p(f)\} = 0.$$

Hence this chain algebra map cannot transport a Poisson or Batalin–Vilkovisky operation merely by forgetting its source.

1.7.2 Reduction of a specified double complex

Two odd differentials require their anticommutator to vanish on their common domain. Their separate nilpotence does not imply this equation. For example, on $\Lambda(x, y)$, exterior multiplication $d_1 = x \wedge -$ and left differentiation $d_2 = \partial_x$ are odd and square zero, but $d_1 d_2 + d_2 d_1 = 1$.

THEOREM 1.7.2 (*A comparison under a second differential*). Let $X^{p,q}, Y^{p,q}$ be bigraded complex vector spaces, zero unless $0 \leq p \leq N$. Let d have bidegree $(1, 0)$ and δ have bidegree $(0, 1)$, with

$$d^2 = \delta^2 = 0, \quad d\delta + \delta d = 0.$$

Suppose $F : X \rightarrow Y$ has bidegree $(0, 0)$, commutes with both differentials, and induces a quasi-isomorphism on every column $(X^{p,*}, \delta)$. Then

$$\text{Tot}(F) : \left(\bigoplus_{p+q=*} X^{p,q}, d + \delta \right) \longrightarrow \left(\bigoplus_{p+q=*} Y^{p,q}, d + \delta \right)$$

is a quasi-isomorphism. The formula is $\text{Tot}(F)((x^{p,q})) = (F(x^{p,q}))$.

Proof. Filter the total complexes decreasingly by the first degree p . This filtration is finite, and the differential on each quotient is δ . The same filtration on the mapping cone has acyclic quotients by hypothesis. The short exact sequence of two successive filtration stages gives a long exact sequence in cohomology. Induction through the finite filtration proves that the cone is acyclic. The two commutation identities give the total chain identity before any cohomology is taken. $\square$

For quantum Drinfeld–Sokolov reduction, the input category in Arakawa’s vanishing theorem is KL_k . It consists of the specified graded affine modules with locally finite action of the finite-dimensional Lie algebra. The theorem gives $H_f^j(M) = 0$ for $j \neq 0$ and exactness of H_f^0 on this category [1, §6, p. 585; Theorem 7.1(i), p. 588]. Applying it to a collision complex requires an affine action on each term, membership in this category, and equivariance of all collision maps. To use the displayed totalization, one also needs its boundedness hypothesis, or a proved substitute for convergence. The nonlinear $\mathcal{W}$ -algebra and a conformal grading alone do not specify a second semi-infinite differential.

The polynomial maps (1.22)–(1.23) construct the associative comparison for the classical $\mathfrak{sl}_2$ constraint. A chiral comparison additionally needs maps from its collision sheaves, including the vacuum terms of every operator product. The augmentation obstruction in Proposition 1.6.1 excludes the augmentation-kernel construction for the matter–ghost vertex algebra.

1.7.3 The Virasoro target at a reduced level

COROLLARY 1.7.3 (*Nilpotence on the Virasoro vacuum carrier*). Let $k \in \mathbb{C} \setminus \{-2\}$ and let $M_{c(k)}$ be the universal Virasoro vacuum module with

$$c(k) = 1 - \frac{6(k+1)^2}{k+2}.$$

Use the ghost module and charge Q of Theorem 1.3.2 on $M_{c(k)} \otimes \mathcal{F}$. This charge squares to zero on the full algebraic carrier if and only if

$$k = -\frac{8}{3} \quad \text{or} \quad k = -\frac{7}{2}.$$

At either level, its absolute vacuum cohomology has one-dimensional components in degrees 0, 3. Its relative vacuum cohomology has one-dimensional components in degrees 0, 2. All other components vanish.

Proof. Theorem 1.3.2 gives

$$Q^2 = \frac{c(k) - 26}{12} \sum_{n>0} (n^3 - n) c_{-n} c_n.$$

On $\mathbf{1} \otimes b_{-2}\Omega$ the output is $(c(k) - 26)\mathbf{1} \otimes c_{-2}\Omega/2$, a nonzero vector when $c(k) \neq 26$. The equation $c(k) = 26$ is $6k^2 + 37k + 56 = (3k+8)(2k+7) = 0$. At these levels the matter module is the universal vacuum module of central charge 26. The four-state calculation and the nonzero-weight contraction in Proposition 1.6.2 compute its absolute and relative cohomology. $\square$

Thus the full displayed ghost carrier cannot define semi-infinite cohomology at arbitrary $k \neq -2$. Another semi-infinite construction would need another explicitly defined differential and a comparison with this one at the applicable parameters. The vacuum PBW character $\prod_{n \geq 2} (1 - q^n)^{-1}$ counts states by conformal weight. It does not compute the cohomology in the preceding corollary.

1.8 CENTRAL COEFFICIENTS AND MAPS OVER CURVES

1.8.1 Level reflection and central charge

PROPOSITION 1.8.1 (*Affine scalar normalizations*). Fix nonzero h and a positive integer d . For $k \neq -h$, set

$$k' = -k - 2h, \quad a(k) = \frac{(k+h)d}{2h}, \quad c_{\text{Sug}}(k) = \frac{kd}{k+h}.$$

Then

$$a(k) + a(k') = 0, \quad c_{\text{Sug}}(k) + c_{\text{Sug}}(k') = 2d.$$

For an affine Lie algebra one takes $h = h^\vee$ and $d = \dim \mathfrak{g}$. The half-central-charge normalization has sum d , rather than zero.

Proof. Write $t = k + h \neq 0$. Reflection sends t to $-t$. The two expressions are $a = td/(2h)$ and $c_{\text{Sug}} = d - hd/t$. Their reflected sums give the formulas. $\square$

PROPOSITION 1.8.2 (*Principal central-charge reflection*). Fix rank r and vectors $\rho, \rho^\vee$ in a real inner-product space. For $t \in \mathbb{C}^*$ define

$$c(t) = r - 12(t\|\rho^\vee\|^2 - 2(\rho, \rho^\vee) + t^{-1}\|\rho\|^2).$$

This is the principal central-charge normalization with $t = k + h^\vee$ and long roots of square length 2. For a fixed scalar v , set $\kappa(t) = vc(t)$. Then

$$c(t) + c(-t) = 2r + 48(\rho, \rho^\vee), \quad \kappa(t) + \kappa(-t) = v(2r + 48(\rho, \rho^\vee)).$$

For $\mathfrak{sl}_2$, $c(t) = 13 - 6(t + t^{-1})$ and $v = 1/2$, so the sums are 26 and 13. For $\mathfrak{sl}_3$, $c(t) = 50 - 24(t + t^{-1})$ and $v = 5/6$, so the sums are 100 and $250/3$.

Proof. The t and t^{-1} terms cancel under reflection. For $\mathfrak{sl}_2$, the positive root α satisfies $(\alpha, \alpha) = 2$, and $\rho = \rho^\vee = \alpha/2$ has square $1/2$. For $\mathfrak{sl}_3$, with simple roots α_1, α_2 of square 2 and $(\alpha_1, \alpha_2) = -1$, one has $\rho = \rho^\vee = \alpha_1 + \alpha_2$ of square 2. Substitution proves the two cases. $\square$

The parameter reflection $t \mapsto -t$ differs from Feigin–Frenkel duality, whose dual parameters satisfy $t t^\vee = 1/r^\vee$, where $r^\vee$ is the lacing number [1, p. 566, footnote 1]. For example, $k = -8/3$ reflects to $k' = -4/3$ in $\mathfrak{sl}_2$. Their central charges are 26, 0, and their half-central charges sum to 13. String ghosts contribute central charge -26 , or half-central charge -13 . They cancel the reflected pair's total central charge. For a single matter module, Theorem 1.3.2 instead gives the operator coefficient $(C - 26)/12$. These are distinct scalar normalizations attached to specified expressions.

1.8.2 The equation for transport of an anomaly operator

PROPOSITION 1.8.3 (*Transport of a square*). Let d_X, d_Y be degree-one operators on graded vector spaces and $F : X \rightarrow Y$ a degree-zero map. If $d_Y F = F d_X$, then

$$d_Y^2 F = F d_X^2.$$

In particular, a chain isomorphism between square-zero complexes supplies no identification with a nonzero anomaly operator on another carrier. If F is injective and $d_Y^2 = 0$, its intertwining equation forces $d_X^2 = 0$.

Proof. Apply d_Y to the intertwining equation and use that equation once more. For an injective F , the resulting identity $F d_X^2 = 0$ implies $d_X^2 = 0$. $\square$

PROPOSITION 1.8.4 (*Horizontal comparison*). Let X be a smooth manifold. Let $(E^\bullet, d_E, \nabla_E)$ and $(F^\bullet, d_F, \nabla_F)$ be bounded complexes of finite-rank complex vector bundles with flat connections. Assume d_E, d_F commute with the connections. On $\Omega^p(X, E^q)$ use total differential $\nabla_E + (-1)^p d_E$, and use the same convention for F . A degree-zero bundle map $\phi : E^\bullet \rightarrow F^\bullet$ induces a map of these total complexes if and only if

$$d_F \phi = \phi d_E, \quad \nabla_F \phi = (\mathbf{1} \otimes \phi) \nabla_E.$$

If these identities hold and ϕ is a quasi-isomorphism on each fiber, it induces an isomorphism on the hypercohomology of the two local-system complexes.

Proof. On a section of form degree zero, the two components of the total chain equation lie in different form degrees. They give exactly the displayed identities. The Leibniz rule extends these identities to arbitrary forms. Flat local frames identify each stalk of the kernel local systems with its fiber. The cone of the induced local-system map is therefore acyclic on every stalk. Its cohomology sheaves vanish, so it is zero in the derived category of sheaves. Applying derived global sections gives the asserted hypercohomology isomorphism. $\square$

Example 1.8.5 (*A genus-one obstruction to fiberwise transport*). On the oriented torus $T^2 = \mathbb{R}^2 / \mathbb{Z}^2$, let L_λ be the rank-one complex local system with monodromy $\lambda \in \mathbb{C}^*$ on the first circle and $\mathbf{1}$ on the second. Its cellular cochain complex, with the product orientation, is

$$(\Lambda^\bullet \mathbb{C}^2, (\lambda - \mathbf{1})e_1 \wedge -).$$

This is the tensor product of the two circle complexes with differentials $\lambda - \mathbf{1}$ and $\mathbf{0}$. If $\lambda \neq \mathbf{1}$, contraction by the dual vector e_1^* , divided by $\lambda - \mathbf{1}$, gives $dh + hd = \mathbf{1}$. Thus $H^*(T^2, L_\lambda) = \mathbf{0}$. For $\lambda = \mathbf{1}$, the differential is zero and the dimensions are $(\mathbf{1}, 2, \mathbf{1})$. The fibers in both cases are one-dimensional complexes concentrated in degree zero. No fiberwise dimension or scalar central coefficient determines the genus-one cohomology.

A comparison over the moduli of curves needs maps of its sheaves or local systems, compatible with the connections and with clutching at the boundary. The genus-zero filtered lifting criterion alone does not define such maps. The same requirement applies separately to pairings, anomaly operators, and the products used in a Maurer–Cartan equation.

1.9 INSERTION INTEGRALS AND FLATNESS EQUATIONS

1.9.1 Marked curves and insertion dependence

For $n \geq 3$, let $C_n(\mathbb{P}^1)$ be the space of ordered distinct points on the complex projective line. The diagonal action of $\mathrm{PGL}_2(\mathbb{C})$ is free. There is a unique projective transformation taking the first three points to $\mathbf{0}, \mathbf{1}, \infty$. The remaining points give the explicit quotient

$$C_n(\mathbb{P}^1) / \mathrm{PGL}_2(\mathbb{C}) \cong \mathcal{M}_{\mathbf{0}, n}.$$

Thus the two spaces have complex dimensions n and $n - 3$. Their compactifications retain these dimensions. In particular, a compactification of $C_n(\mathbb{P}^1)$ cannot equal $\overline{\mathcal{M}}_{\mathbf{0}, n+1}$, whose dimension is $n - 2$.

PROPOSITION 1.9.1 (*Insertion integrals and Euler characteristics*). Put $Z = \overline{\mathcal{M}}_{0,n}(\mathbb{C})$ with its complex orientation and real dimension $m = 2n - 6$. Let $C_1, \dots, C_n$ be complex cochain complexes. An insertion map is a degree-zero chain map

$$\mu : C_1 \otimes \cdots \otimes C_n \longrightarrow \Omega^\bullet(Z; \mathbb{C}),$$

where the tensor differential has sign $(-1)^{|x_1| + \cdots + |x_{j-1}|}$ on its j th summand. For closed homogeneous inputs of total degree m , define

$$\mathcal{A}_\mu(x_1, \dots, x_n) = \int_Z \mu(x_1 \otimes \cdots \otimes x_n).$$

This is a multilinear function of their cohomology classes. If, on a fixed product of input spaces, it equals an input-independent scalar for every tuple, that scalar is zero. Therefore an identity with an Euler characteristic of a fixed complex requires separate insertion normalization, and cannot hold for arbitrary insertions when that Euler characteristic is nonzero.

Proof. The chain identity for μ sends a tensor boundary to an exact form. Stokes' theorem on the compact oriented manifold Z makes the integral of such a form zero. Multilinearity follows from the linearity of μ and integration. Putting one input equal to zero forces the scalar to be zero. For an explicit instance, take $n = 3$, Z a point, and $C_j = \mathbb{C}$ in degree zero. The multiplication map gives $\mathcal{A}_\mu(a, b, c) = abc$. The Euler characteristic of the degree-zero complex $\mathbb{C}$ is 1. The tuples $(1, 1, 1)$ and $(0, 1, 1)$ give different amplitudes for the same complex. $\square$

For a chiral-chain application, μ includes the actual insertion and collision maps into forms on Z . Logarithmic forms along a divisor require an integration domain and boundary prescription for which the integration functional is closed. An Euler class integral supplies a specific characteristic form, rather than an arbitrary multilinear insertion.

1.9.2 The differential graded Lie algebra of connections

THEOREM 1.9.2 (*Flatness and the Chern–Simons functional*). Let M be a closed oriented smooth three-manifold and $\mathfrak{g}$ a finite-dimensional complex Lie algebra with a nondegenerate invariant symmetric bilinear form $\langle -, - \rangle$. Consider connections on the trivial bundle, written as $A \in \Omega^1(M; \mathfrak{g})$. On $L = \Omega^\bullet(M) \otimes \mathfrak{g}$ use differential d_{dR} and bracket

$$[\alpha \otimes x, \beta \otimes y] = (\alpha \wedge \beta) \otimes [x, y].$$

For $k \in \mathbb{C}^*$, put

$$S_k(A) = \frac{k}{4\pi} \int_M \left(\langle A \wedge dA \rangle + \frac{1}{3} \langle A \wedge [A, A] \rangle \right).$$

Its critical points are exactly the Maurer–Cartan elements of L :

$$F_A = dA + \frac{1}{2} [A, A] = 0.$$

Proof. The de Rham Leibniz rule and the Lie Jacobi identity make L a differential graded Lie algebra. Its Maurer–Cartan equation in degree one is the displayed equation by definition. For an arbitrary variation $a \in \Omega^1(M; \mathfrak{g})$, integration by parts on the closed manifold gives

$$\left. \frac{d}{dt} \right|_{t=0} \int_M \langle (A + ta) \wedge d(A + ta) \rangle = 2 \int_M \langle a \wedge dA \rangle.$$

Invariance of the bilinear form makes the three differentiated factors of the cubic term equal after the graded signs are included. Its variation is therefore $\int_M \langle a \wedge [A, A] \rangle$. Consequently

$$(dS_k)_A(a) = \frac{k}{2\pi} \int_M \langle a \wedge F_A \rangle.$$

If this vanishes for all smooth variations supported in a coordinate chart, nondegeneracy of the form forces every coefficient of F_A to vanish. The converse follows from the same formula. $\square$

This construction uses $\Omega^\bullet(M) \otimes \mathfrak{g}$. The Chevalley–Eilenberg cochains $\text{Hom}(\Lambda^\bullet \mathfrak{g}, \mathbb{C})$ are a different complex. Transport from a chiral bar object to this differential graded Lie algebra requires the corresponding bracket data or higher morphism components.

1.9.3 Transport of Maurer–Cartan elements

PROPOSITION 1.9.3 (*The components required for Maurer–Cartan transport*). Let L, L' be uncurved L_∞ algebras over $\mathbb{C}$. In the symmetric convention, these are graded vector spaces with degree-one square-zero coderivations Q, Q' on the coaugmented symmetric coalgebras of $L[1], L'[1]$, vanishing on the coaugmentation. Let $\mathfrak{m}$ be the nilpotent maximal ideal of a commutative local Artinian complex algebra. Suppose a coaugmentation-preserving coalgebra map F satisfies $Q'F = FQ$. Writing its Taylor components as $F_j : S^j(L[1]) \rightarrow L'[1]$, define, for $\alpha \in (L \otimes \mathfrak{m})^1$,

$$s\beta = \sum_{j \geq 1} \frac{1}{j!} F_j((s\alpha)^{\odot j}).$$

Then α Maurer–Cartan implies β Maurer–Cartan. A linear chain quasi-isomorphism without these coalgebra equations need not have this property.

Proof. Nilpotence of $\mathfrak{m}$ makes every displayed sum finite. Set $v = s\alpha$ of degree zero and use $e^v = \sum_{j \geq 0} v^{\odot j} / j!$. The coderivation identity gives

$$Q(e^v) = e^v \odot \left(\sum_{j \geq 1} \frac{1}{j!} Q_j(v^{\odot j}) \right).$$

This can be checked by selecting the j inputs on which Q_j acts. The binomial coefficient cancels the factorials. The expression in parentheses is the shifted Maurer–Cartan expression. The coalgebra identity similarly gives $F(e^v) = e^{s\beta}$. Thus $Q(e^v) = 0$ implies $Q'(e^{s\beta}) = Q'F(e^v) = FQ(e^v) = 0$. Projection to $L'[1]$ proves the assertion.

For the last statement, let both underlying complexes have basis x in degree one and y in degree two, with zero differential. Give the first algebra zero brackets, and the second the sole nonzero bracket $[x, x] = y$. Graded skew symmetry permits this bracket because x is odd, and Jacobi holds because y is central. The identity of complexes is a quasi-isomorphism. Over $\mathbb{C}[t]/(t^3)$, tx is Maurer–Cartan in the first algebra but has curvature $t^2 y/2 \neq 0$ in the second. $\square$

COROLLARY 1.9.4 (*Flatness on an identified carrier*). In the preceding proposition, suppose the target is $L' = \Omega^\bullet(M) \otimes \mathfrak{g}$ from Theorem 1.9.2. Then the constructed β is a flat connection with coefficients in $\mathfrak{m}$:

$$d\beta + \frac{1}{2}[\beta, \beta] = 0.$$

The higher Jacobi identities alone do not make every degree-one element Maurer–Cartan.

Proof. The first assertion is the preceding proposition and the definition of curvature on the target. For the second, the abelian differential graded Lie algebra with $L^1 = \mathbb{C}x$, $L^2 = \mathbb{C}y$, and $dx = y$ satisfies all Jacobi identities. The element tx over $\mathbb{C}[t]/(t^2)$ has nonzero Maurer–Cartan expression ty . $\square$

Reduction to a Drinfeld–Sokolov constraint requires the reduction maps and their compatibility with these higher components. The polynomial reduction of Theorem 1.7.1 supplies a chain algebra comparison on its polynomial carrier. Its projection fails Poisson compatibility, so it does not by itself give this transport of flatness equations.

1.10 CHARACTERS AND ELLIPTIC KERNELS

PROPOSITION 1.10.1 (*Translation of an integrally graded character*). Let $V = \bigoplus_{n \geq 0} V_n$ have finite-dimensional components and let $C \in \mathbb{C}$. Suppose $\sum_n \dim(V_n)q^n$ converges for $|q| < 1$. For $\tau \in \mathbb{H}$ put

$$Z_V(\tau) = \sum_{n \geq 0} \dim(V_n) \exp(2\pi i \tau (n - C/24)).$$

Then $Z_V(\tau + 1) = e^{-2\pi i C/24} Z_V(\tau)$. For two such nonzero characters of real central charges $C, \tilde{C}$, the product $Z_V \overline{Z_{\tilde{V}}}$ is invariant under translation if and only if $C - \tilde{C} \in 24\mathbb{Z}$. This translation condition alone does not imply invariance under all modular transformations.

Proof. Each summand acquires the factor $e^{2\pi i n} e^{-2\pi i C/24} = e^{-2\pi i C/24}$. Absolute convergence permits the factor to be taken outside the sum. The conjugate character supplies the inverse phase with $\tilde{C}$, giving the stated condition. Translation is only one generator of the modular group. For example $e^{2\pi i \tau}$ is translation invariant but has different values at $2i$ and $i/2$. $\square$

THEOREM 1.10.2 (*The Heisenberg vacuum character*). The rank-one Heisenberg vacuum module has character

$$Z_H(\tau) = e^{-\pi i \tau / 12} \prod_{n \geq 1} (1 - e^{2\pi i n \tau})^{-1} = \eta(\tau)^{-1}.$$

It satisfies

$$Z_H(\tau + 1) = e^{-\pi i / 12} Z_H(\tau), \quad Z_H(-1/\tau) = (-i\tau)^{-1/2} Z_H(\tau).$$

The square root is the holomorphic branch positive when τ is positive imaginary. The modular weight of this character is $-1/2$, with the inverse eta multiplier.

Proof. A vacuum basis consists of $\prod_{n \geq 1} a_{-n}^{m_n} \mathbf{1}$, with finitely many nonzero $m_n \geq 0$ and conformal weight $\sum n m_n$. Summing the geometric series for each oscillator gives the displayed product. It converges normally on compact subsets of $\mathbb{H}$ because $\sum |q|^n / (1 - |q|^n)$ converges there. The vacuum energy is $-1/24$ since the central charge is one. The translation identity follows directly from the product. The eta transformation formula gives $\eta(-1/\tau) = (-i\tau)^{1/2} \eta(\tau)$ [2, Eq. 23.18.5]. Taking reciprocals proves the second identity with the same branch. $\square$

For the universal Virasoro vacuum, replace the oscillator product by $\prod_{n \geq 2} (1 - q^n)^{-1}$ and the exponent by $-C/24$. Its translation multiplier remains $e^{-2\pi i C/24}$. For an integrally graded affine vacuum with its Sugawara conformal vector, that multiplier uses $C = k \dim \mathfrak{g} / (k + h^\vee)$. Neither computation asserts a one-dimensional modular representation for the whole character space.

PROPOSITION 1.10.3 (*Invariant pairings of character vectors*). Let $\chi : \mathbb{H} \rightarrow \mathbb{C}^r$ satisfy $\chi(\gamma\tau) = \rho(\gamma)\chi(\tau)$ for a specified representation ρ of $SL_2(\mathbb{Z})$. If a constant Hermitian matrix H satisfies $\rho(\gamma)^* H \rho(\gamma) = H$ for every γ , then $\chi(\tau)^* H \chi(\tau)$ is modular invariant.

Proof. Substitution gives $\chi(\gamma\tau)^* H \chi(\gamma\tau) = \chi(\tau)^* \rho(\gamma)^* H \rho(\gamma) \chi(\tau) = \chi(\tau)^* H \chi(\tau)$. $\square$

1.10.1 The Laurent normalization of a meromorphic kernel

For $\Lambda_\tau = \mathbb{Z} + \tau\mathbb{Z}$, define

$$\zeta_\tau(z) = \frac{1}{z} + \sum_{\omega \in \Lambda_\tau \setminus \{0\}} \left(\frac{1}{z - \omega} + \frac{1}{\omega} + \frac{z}{\omega^2} \right), \quad z \notin \Lambda_\tau.$$

The grouped summand is $O(|\omega|^{-3})$ on compact subsets away from the poles, so the series converges normally. Set $E_2 = (12/\pi i) \partial_\tau \log \eta$ and $G_\tau(z) = \zeta_\tau(z) + (\pi^2/3) E_2(\tau) z$.

PROPOSITION 1.10.4 (*Transformation of the normalized kernel*). For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$, put $j = c\tau + d$. Then

$$G_{\gamma\tau}(z/j) = j G_\tau(z) - 2\pi i c z.$$

Both sides have residue j as functions of z at zero. The pulled-back one-form satisfies

$$G_{\gamma\tau}(z/j) d(z/j) = G_\tau(z) dz - \frac{2\pi i c}{j} z dz,$$

where the differential is in the fiber coordinate z with τ fixed.

Proof. The identity $\Lambda_{\gamma\tau} = j^{-1}\Lambda_\tau$ and reindexing of the normally convergent series give $\zeta_{\gamma\tau}(z/j) = j \zeta_\tau(z)$. The eta multiplier is independent of τ . Logarithmic differentiation of its transformation law [2, Eq. 23.18.5] therefore gives

$$E_2(\gamma\tau) = j^2 E_2(\tau) + \frac{6c j}{\pi i}.$$

For $c = 0$ the same identity follows directly from translation and the central element $-I$. Substitution proves the kernel identity because $2\pi/i = -2\pi i$. The leading term $1/z$ proves the residue assertion, and division by j gives the one-form identity. $\square$

The function G_τ is a meromorphic expression on the covering plane with the stated transformation law. A propagator on an elliptic curve requires its additional periodicity, coefficient bundle, and differential equation. The character, this kernel, and a chiral chain complex are separate objects.

Conjecture 1.10.5 (*Chiral realization of modular transformation laws*). For a conformal chiral algebra with a specified convergent character space and a compatible family of elliptic chiral chain complexes, there is a modular-equivariant chain-level realization of its character transformation law. Such a realization includes descent maps on the family of elliptic curves and a trace map compatible with those descent maps. The scalar character identities above do not construct these maps.

1.11 REDUCTION PARAMETERS AND FINITE BATALIN–VILKOVISKY COMPLEXES

PROPOSITION 1.11.1 (*Conformal obstruction to parameter reflection*). Let $t \in \mathbb{C}^*$ and put $c(t) = 13 - 6(t + t^{-1})$. A conformal vertex algebra isomorphism between the universal Virasoro algebras at charges $c(t)$ and $c(-t)$ exists if and only if $t = \pm i$.

Proof. A conformal isomorphism preserves the vacuum and conformal vector T . It therefore preserves $T_{(3)}T = (c/2)\mathbf{1}$, so the two charges must agree. Since their sum is 26, equality is equivalent to $c(t) = 13$, or $t + t^{-1} = 0$. This gives $t = \pm i$. Conversely both universal algebras then have charge 13, and the map sending their conformal generators to one another is an isomorphism by their identical universal presentation. $\square$

At $t = -2/3$, reflection gives $c(t) = 26$ and $c(-t) = 0$, whereas the reciprocal partner $t^{-1} = -3/2$ has charge 26. This distinguishes reflection from conformal Feigin–Frenkel duality by an actual vertex product. It does not exclude a curved or nonconformal comparison with different structure-preservation requirements.

Example 1.11.2 (*The nonsimply laced central coefficient*). For B_2 , take an orthonormal basis e_1, e_2 and positive roots $e_1 - e_2, e_2, e_1, e_1 + e_2$. The coroot of α is $2\alpha/(\alpha, \alpha)$. Thus

$$\rho = (3/2, 1/2), \quad \rho^\vee = (2, 1), \quad (\rho, \rho^\vee) = 7/2.$$

The principal charge is $c(t) = 86 - 60t - 30/t$, so $c(t) + c(-t) = 172$. The rank is 2, the dimension is $2 + 2 \cdot 4 = 10$, and $h^\vee = 1 + (\rho, e_1 + e_2) = 3$. Consequently $2r + 4h^\vee \dim \mathfrak{g} = 124$ cannot be the reflected sum for all simple Lie algebras. For simply laced root systems $\rho = \rho^\vee$; the general expression remains $2r + 48(\rho, \rho^\vee)$.

PROPOSITION 1.11.3 (*Two genus-one scalar normalizations*). Let $B_2 = 1/6$ be the second Bernoulli number and define the scalar coefficient

$$F_1(\kappa) = \kappa \frac{2^1 - 1}{2^1} \frac{|B_2|}{2!}.$$

Then $F_1(\kappa) = \kappa/24$. Under $\kappa = C/2$, this is $C/48$. The vacuum-energy exponent in $q^{L_0 - C/24}$ is instead $-C/24$. At $C = 26$ these values are $13/24$ and $-13/12$.

Proof. The Bernoulli coefficient is $(1/2)(1/6)(1/2) = 1/24$. Substitution gives the other values. $\square$

This proposition concerns the specified Bernoulli coefficient. Its identification with a Hodge integral or a trace requires that map and its normalization. The vacuum energy uses the separately defined conformal trace.

THEOREM 1.11.4 (*A polynomial Batalin–Vilkovisky differential and its quotient*). Let $R = \mathbb{C}[x_1, \dots, x_n]$ and $S \in R$. Give $K = R \otimes \Lambda(\eta_1, \dots, \eta_n)$ degrees $|x_i| = 0, |\eta_i| = -1$. Use the degree-one odd Poisson bracket with $\{x_i, \eta_j\} = \delta_{ij}, \{x_i, x_j\} = \{\eta_i, \eta_j\} = 0$, extended by graded skew symmetry and the graded Leibniz rules. Then $Q = \{S, -\}$ is the Koszul differential

$$Qx_i = 0, \quad Q\eta_i = \partial_i S,$$

and satisfies $Q^2 = 0$. If $\partial_1 S, \dots, \partial_n S$ is a regular sequence, the chain algebra projection

$$p : K \longrightarrow R/(\partial_1 S, \dots, \partial_n S), \quad p(\eta_i) = 0$$

is a quasi-isomorphism.

Proof. The bracket formulas give the displayed derivation. Its square vanishes on every generator and hence everywhere. The projection is a chain map because it kills each $Q\eta_i$. For a single regular element f , the complex $R\eta \xrightarrow{f} R$ has zero cohomology in degree -1 and quotient $R/(f)$ in degree zero. Inductively tensor with the next two-term complex. The two-column spectral sequence first computes the preceding Koszul cohomology, then multiplication by the next element on the quotient. Regularity makes that multiplication injective. The filtration is finite, so induction proves concentration in degree zero and the asserted projection isomorphism. $\square$

For $S = \frac{1}{2} \sum_i x_i^2$, the target is $\mathbb{C}$ and the projection evaluates $x_i = \eta_i = 0$. This is an explicit finite polynomial BRST comparison. It has no collision sheaves, ultraviolet completion, or curvewise descent. A comparison with chiral chains must construct those additional objects and maps.

1.12 FORM DEGREE IN HIGHER INTERACTION TERMS

PROPOSITION 1.12.1 (*Quartic expressions on a three-manifold*). Let M be a smooth three-manifold and $A \in \Omega^1(M; W)$ for a finite-dimensional vector space W . Any expression obtained by applying a four-linear map to coefficients of A and wedging all four one-form factors is zero. If a vector field v and a four-linear map $T : W^{\otimes 4} \rightarrow \mathbb{C}$ are specified, the expression

$$T(\iota_v A, A, A, A) \in \Omega^3(M)$$

is instead a defined three-form. It can be nonzero, but depends on the additional contraction and tensor.

Proof. The first expression belongs to $\Omega^4(M) = 0$. For the second, $\iota_v A$ has form degree zero, so the total degree is three. For an explicit local example take $W = \mathbb{C}^3$, $A = e_1 dx + e_2 dy + e_3 dz$, $v = \partial_x$, and the tensor whose only nonzero coordinate entry is $T(e_1, e_1, e_2, e_3) = 1$. Then the expression is $dx \wedge dy \wedge dz$. $\square$

An RTT relation is an algebraic relation on matrix-valued series. To use it as an action term, a map to a three-form such as the displayed contraction must be defined and its required symmetries proved. A critical affine level alone does not make the Lie bracket in $\Omega^\bullet(M) \otimes \mathfrak{g}$ vanish. For example, $A = e dx + f dy$ in $\mathfrak{sl}_2$ has $[A, A]/2 = h dx \wedge dy \neq 0$, independently of any level parameter.

PROPOSITION 1.12.2 (*Integration on a finite quotient*). Let a finite group Γ act freely by orientation-preserving diffeomorphisms on a closed oriented three-manifold M . Let ω be an invariant three-form and $\bar{\omega}$ its descended form on M/Γ . Then

$$\int_{M/\Gamma} \bar{\omega} = \frac{1}{|\Gamma|} \int_M \omega.$$

Proof. The quotient map is an oriented covering of degree $|\Gamma|$. An evenly covered coordinate cover and a partition of unity reduce integration to the sum over its $|\Gamma|$ sheets. Each sheet has the same integral, giving the formula. $\square$

For a nonfree finite action, integration on the quotient stack uses the right-hand side as its normalization. An additional division by $|\Gamma|$ after taking this quotient integral changes the normalization. Neither a fractional grading nor a W -algebra specifies an action of Γ on M . A pole of residue one cannot be the only singularity of a meromorphic differential on a compact torus. A Cauchy kernel therefore requires a correction that removes the constant mode. The correction below is smooth away from the pole, respects changes of lattice basis, and gives an explicit homotopy for the Dolbeault differential.

1.13 THE PERIODIC KERNEL

Fix $\tau \in \mathbb{H}$, put $Y = \text{Im } \tau$, and let $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$. Write $z = x + iy$, orient the torus by $dx \wedge dy$, and use the normalized area form

$$\nu_\tau = \frac{dx \wedge dy}{Y} = \frac{i}{2Y} dz \wedge d\bar{z}, \quad \int_{E_\tau} \nu_\tau = 1.$$

The Dirac current δ_o is the degree-two current with $\langle \delta_o, \varphi \rangle = \varphi(o)$ for a smooth function φ . Our convention is $\bar{\partial}f = (\partial_{\bar{z}}f)d\bar{z}$, where $\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$.

For the lattice $\Lambda_\tau = \mathbb{Z} + \tau\mathbb{Z}$, define

$$\zeta_\tau(z) = \frac{1}{z} + \sum_{\omega \in \Lambda_\tau \setminus \{0\}} \left(\frac{1}{z - \omega} + \frac{1}{\omega} + \frac{z}{\omega^2} \right).$$

The grouped summands are $O(|\omega|^{-3})$ on compact subsets away from the poles. Thus the series and its termwise derivatives converge normally there. Pairing ω and $-\omega$ shows that ζ_τ is odd and has expansion $z^{-1} + O(z^3)$ at zero. This is the lattice normalization of [7, Eq. 23.2.5].

LEMMA 1.13.1. There are constants a_τ, b_τ with

$$\zeta_\tau(z+1) - \zeta_\tau(z) = a_\tau, \quad \zeta_\tau(z+\tau) - \zeta_\tau(z) = b_\tau, \quad a_\tau\tau - b_\tau = 2\pi i.$$

Proof. Put $\wp_\tau = -\zeta'_\tau$. Its derivative is the absolutely normally convergent series $-2 \sum_{\omega \in \Lambda_\tau} (z - \omega)^{-3}$. Reindexing proves that this derivative is periodic. Hence $\wp_\tau(z + \lambda) - \wp_\tau(z)$ is constant for $\lambda = 1, \tau$. The function $\wp_\tau$ is even; evaluating the difference at $z = -\lambda/2$ shows that the constant is zero. The half-period is not a lattice point for these primitive periods. Differentiating the two displayed differences of ζ_τ now gives zero, proving constancy.

Choose a translated fundamental parallelogram with no pole on its boundary. It contains one pole of residue one. The residue theorem gives $\oint \zeta_\tau(z) dz = 2\pi i$. Pairing its opposite sides gives $a_\tau\tau - b_\tau$, with the horizontal sides contributing $-b_\tau$ and the sides parallel to τ contributing $a_\tau\tau$. This proves the last identity. $\square$

THEOREM 1.13.2. The function

$$P_\tau(z) = \zeta_\tau(z) - a_\tau z + \frac{2\pi i}{Y} \text{Im } z \tag{1.24}$$

is odd and Λ_τ -periodic. It is smooth away from the lattice, locally integrable, and has singular part $1/z$. The one-form $P_\tau(z)dz$ therefore defines a current on E_τ and satisfies

$$\bar{\partial}(P_\tau(z)dz) = 2\pi i(\delta_o - \nu_\tau). \tag{1.25}$$

Among odd periodic functions with singular part $1/z$ and smooth remainder, it is the unique solution of this current equation.

Proof. Translation by one changes the first two terms of (1.24) by $a_\tau - a_\tau = 0$ and leaves the last term unchanged. Translation by τ changes the expression by $b_\tau - a_\tau\tau + 2\pi i = 0$. Each term is odd. The only singularity in a fundamental domain is $1/z$, which is locally integrable because $\int_0^\epsilon r^{-1}r dr < \infty$.

Away from the poles,

$$\partial_{\bar{z}}P_\tau = -\frac{\pi}{Y}.$$

At a pole, the distribution identity $\partial_{\bar{z}}(1/z) = \pi \delta_0^{\text{area}}$ uses Dirac mass relative to $dx dy$. To fix its sign and coefficient, apply Stokes' theorem to a test function on the complement of a disk of radius ϵ . The inner boundary has clockwise orientation, while the distributional derivative contributes the opposite boundary term. The resulting positively oriented circle integral tends to $\varphi(o) \oint dz/z = 2\pi i \varphi(o)$. Thus $\bar{\partial}(dz/z) = 2\pi i \delta_0$ in current notation. For the smooth term,

$$-\frac{\pi}{Y} d\bar{z} \wedge dz = \frac{\pi}{Y} dz \wedge d\bar{z} = -2\pi i \nu_\tau.$$

These identities prove (1.25).

The difference of two proposed solutions has a smooth coefficient on the torus, is annihilated by $\partial_{\bar{z}}$, and hence is holomorphic. A holomorphic function on a compact connected torus is constant by the maximum principle. Oddness makes that constant zero. $\square$

COROLLARY 1.13.3 (*The constant mode*). There is no current K of degree one on E_τ with $\bar{\partial}K = 2\pi i \delta_0$. In particular, there is no meromorphic one-form with a unique simple pole of residue one.

Proof. Pair the asserted equation with the constant test function one. The left side is zero because the derivative of that test function is zero. The right side is $2\pi i$. $\square$

1.14 CHANGE OF LATTICE BASIS

THEOREM 1.14.1. Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$, $\tau' = (a\tau + b)/(c\tau + d)$ and $j = c\tau + d$. Then

$$P_{\tau'}(z/j) = j P_\tau(z). \quad (1.26)$$

Consequently the isomorphism $\phi_\gamma : E_\tau \rightarrow E_{\tau'}$ induced by $z \mapsto z/j$ satisfies $\phi_\gamma^*(P_{\tau'} dz') = P_\tau dz$ and $\phi_\gamma^* \nu_{\tau'} = \nu_\tau$.

Proof. The elementary lattice and area identities are $\Lambda_{\tau'} = j^{-1} \Lambda_\tau$ and $\text{Im } \tau' = Y/|j|^2$. Set $R(z) = j^{-1} P_{\tau'}(z/j)$. It is odd and Λ_τ -periodic, and its singular part is $1/z$. Away from the lattice, the chain rule gives

$$\partial_{\bar{z}} R(z) = \frac{1}{j \bar{j}} \left(-\frac{\pi}{\text{Im } \tau'} \right) = -\frac{\pi}{Y}.$$

The residue gives the same delta term as for P_τ . The uniqueness assertion of Theorem 1.13.2 proves $R = P_\tau$. The one-form law follows by $dz' = dz/j$. The area law follows from $dx' dy' = |j|^{-2} dx dy$ and the displayed imaginary-part identity. $\square$

The coefficient of z can also be written in the usual holomorphic normalization. Put $q = e^{2\pi i \tau}$ and

$$E_2(\tau) = 1 - 24 \sum_{n \geq 1} \frac{n q^n}{1 - q^n}.$$

The Fourier expansion of ζ_τ in a neighborhood of zero is

$$\zeta_\tau(z) = a_\tau z + \pi \cot(\pi z) + 4\pi \sum_{n \geq 1} \frac{q^n}{1 - q^n} \sin(2\pi n z).$$

This is [7, Eq. 23.8.2] with half-periods $1/2, \tau/2$. On $|\operatorname{Im} z| \leq Y - \epsilon$ the differentiated series converges normally by geometric decay. Comparison of the linear coefficient with $\zeta_\tau(z) = 1/z + O(z^3)$ gives

$$a_\tau = \frac{\pi^2}{3} E_2(\tau), \quad P_\tau(z) = \zeta_\tau(z) - \frac{\pi^2}{3} E_2(\tau)z + \frac{2\pi i}{Y} \operatorname{Im} z.$$

The nonholomorphic term is part of the periodic kernel. A different meromorphic expression on the covering plane need not satisfy either its periodicity or its current equation.

1.15 A HOMOTOPY FOR THE DOLBEAULT COMPLEX

Use smooth complex-valued forms on the fixed torus and their usual Fréchet topologies. The Dolbeault complex is concentrated in degrees zero and one:

$$\mathcal{D}_\tau^\bullet = [C^\infty(E_\tau) \xrightarrow{\bar{\partial}} C^\infty(E_\tau) d\bar{z}].$$

For a coefficient function f , write $\langle f \rangle = Y^{-1} \int_{E_\tau} f dx dy$. Let $\mathcal{H}_\tau^\bullet = \mathbb{C} \oplus \mathbb{C} d\bar{z}$ with zero differential, let i include the constant coefficients, and let p take coefficientwise averages.

THEOREM 1.15.1. Define a degree-minus-one operator by $H_\tau = 0$ on degree zero and

$$H_\tau(f d\bar{z})(z) = \frac{1}{\pi} \int_{E_\tau} P_\tau(z-w) f(w) dx_w dy_w. \quad (1.27)$$

The integral is absolutely convergent, defines a smooth function, and is independent of representatives of z, w in the covering plane. The maps are continuous and satisfy

$$pi = 1, \quad \bar{\partial} H_\tau + H_\tau \bar{\partial} = 1 - ip, \quad H_\tau i = p H_\tau = H_\tau^2 = 0.$$

They give a deformation retraction of the smooth Dolbeault complex onto its constant coefficient subcomplex. For every lattice-basis change ϕ_γ , the homotopy obeys $\phi_\gamma^* H_{\tau'} = H_\tau \phi_\gamma^*$.

Proof. Periodicity makes the integrand a well-defined function of the difference of two points of the torus. The locally integrable kernel and bounded smooth coefficient give absolute convergence. After changing variables, the integral is $\pi^{-1} \int P_\tau(u) f(z-u) dx_u dy_u$. Derivatives in z may be applied to f under the integral: each derivative is bounded and $P_\tau \in L^1$. This proves smoothness and the seminorm estimate

$$\|H_\tau(f d\bar{z})\|_{C^m} \leq \frac{\|P_\tau\|_{L^1}}{\pi} \|f\|_{C^m}.$$

Thus H_τ is continuous.

The scalar form of (1.25) is $\partial_{\bar{z}} P_\tau = \pi \delta_0^{\text{area}} - \pi/Y$. Convolving it with f yields

$$\bar{\partial} H_\tau(f d\bar{z}) = (f - \langle f \rangle) d\bar{z}.$$

Integration by parts in the distribution sense, or differentiation of the convolution with the derivative placed on f , gives $H_\tau \bar{\partial} f = f - \langle f \rangle$. These are the two degree components of the homotopy identity. The mean of a derivative of a periodic function is zero, so p is a chain map. The inclusion is a chain map and $pi = 1$.

Inversion on the torus preserves area and changes P_τ to its negative. Absolute integrability therefore implies $\int P_\tau = 0$. It follows that $H_\tau i = 0$ and, by Fubini, $p H_\tau = 0$. The square is zero by the two degrees of the complex.

For naturality, write a one-form on $E_{\tau'}$ as $f'(w')d\bar{w}'$. Its pullback has coefficient $f'(w/j)/\bar{j}$. In (1.27), substitute $w = jw'$. The area contributes $|j|^2$, the kernel contributes j^{-1} by (1.26), and the one-form coefficient contributes $\bar{j}^{-1}$. Their product is one, proving the required identity. On degree zero both sides vanish. $\square$

For a smooth $(0, 1)$ -form $f d\bar{z}$ of mean zero, formula (1.27) is the unique mean-zero smooth solution of $\bar{\partial}u = f d\bar{z}$. Existence and normalization follow from the theorem. For uniqueness the difference is holomorphic on the compact torus and hence constant, and its mean is zero. For $f = 1$, the equation has no solution: integration of $\partial_{\bar{z}}u$ over a fundamental domain vanishes.

On the square torus $\tau = i$, the Fourier mode $f(x, y) = e^{2\pi i x}$ satisfies $\partial_{\bar{z}}f = \pi i f$ and has mean zero. Consequently $H_i(f d\bar{z}) = -i f/\pi$. The constant mode is the complementary case: it is precisely the one subtracted in (1.25).

This construction concerns the scalar smooth Dolbeault complex on a compact complex torus. It supplies neither a conformal-character trace on chiral chains nor a comparison with a vertex-algebra BRST complex. Such a comparison must specify its coefficient objects and its compatibility with collision maps and with the displayed contraction. The principal affine $\mathfrak{sl}_2$ constraint admits a closed Virasoro field whose embedding preserves every collision, including the vacuum term. The ghost relation determines the direction of this map.

1.16 CURRENTS, GHOSTS, AND THE DIFFERENTIAL

All vector spaces are over $\mathbb{C}$. Fix $k \in \mathbb{C}$ and put $t = k + 2$. Let $V^k(\mathfrak{sl}_2)$ be the universal affine vacuum vertex algebra with even generators e, h, f and invariant form

$$(e, f) = 1, \quad (h, h) = 2.$$

The nonzero defining λ -brackets are

$$[h_\lambda e] = 2e, \quad [h_\lambda f] = -2f, \quad [e_\lambda f] = h + k\lambda \mathbf{1}, \quad [h_\lambda h] = 2k\lambda \mathbf{1}. \quad (1.28)$$

The other brackets follow by skew symmetry. The affine algebra is universal, including at special levels.

Let F_{bc} be the charged free-fermion vertex superalgebra with odd generators b, c and

$$[b_\lambda c] = [c_\lambda b] = \mathbf{1}, \quad [b_\lambda b] = [c_\lambda c] = 0.$$

Give b, c cohomological degrees $-1, 1$ and the affine generators degree zero. Set $C_k = V^k(\mathfrak{sl}_2) \otimes F_{bc}$, with tensor-product vacuum $\mathbf{1}$ and translation ∂ . For every state a , use

$$Y(a, z) = \sum_{n \in \mathbb{Z}} a_{(n)} z^{-n-1}, \quad [a_\lambda d] = \sum_{n \geq 0} \frac{\lambda^n}{n!} a_{(n)} d, \quad : ad := a_{(-1)} d.$$

Thus $\{b_{(m)}, c_{(n)}\} = \delta_{m+n, -1}$, and all five generating fields have $a_{(n)}\mathbf{1} = 0$ for $n \geq 0$. The carrier consists of finite linear combinations of PBW monomials of negative modes. It has no completion in ghost number, mode number, or tensor length.

Define the odd state and its residue by

$$q =: (e - \mathbf{1})c :, \quad D = q_{(0)} = \text{Res}_{z=0} Y(q, z) dz. \quad (1.29)$$

The residue of $z^{-1}dz$ is 1. The defining operator products give

$$Db = e - \mathbf{1}, \quad Dh = -2 : ec :, \quad Df = : hc : + k \partial c, \quad De = Dc = 0.$$

There are no singular contractions in $Y(q, z)Y(q, w)$. Hence $[q_\lambda q] = 0$, and the zero-mode commutator formula gives $D^2 = 0$. The same formula gives, for every integer n and homogeneous a, d ,

$$D(a_{(n)}d) = (Da)_{(n)}d + (-1)^{|a|}a_{(n)}Dd, \quad D\partial = \partial D. \quad (1.30)$$

This is a differential vertex superalgebra at every level, including $k = -2$. The noncritical hypothesis enters the conformal field below.

1.17 THE CONFORMAL FIELD

Assume $t \neq 0$ and define

$$S = \frac{\frac{1}{2} : hh : + : ef : + : fe :}{2t}, \quad G = : (\partial b)c :, \quad T = S + \frac{1}{2}\partial h + G. \quad (1.31)$$

The conformal modes are $L_n = T_{(n+1)}$.

PROPOSITION 1.17.1. The state T is closed and is a conformal vector on C_k . Its central charge and generating-field weights are

$$C(k) = \frac{3k}{k+2} - 6k - 2 = 13 - 6(t + t^{-1}), \quad (\Delta_e, \Delta_h, \Delta_f, \Delta_b, \Delta_c) = (0, 1, 2, 0, 1).$$

The field q is primary of weight one. In particular, D commutes with every L_n .

Proof. The noncommutative Wick identity is

$$[a_\lambda : uv :] = : [a_\lambda u]v : + (-1)^{p(a)p(u)} : u[a_\lambda v] : + \int_0^\lambda [[a_\lambda u]_\mu v] d\mu,$$

where p denotes parity. Applied to (1.28), it gives

$$[S_\lambda x] = (\partial + \lambda)x \quad (x = e, h, f), \quad [S_\lambda S] = (\partial + 2\lambda)S + \frac{k}{4t}\lambda^3 \mathbf{1}.$$

These are the Sugawara identities with the displayed invariant form. In the ghost factor, the same identity gives

$$[G_\lambda b] = \partial b, \quad [G_\lambda c] = (\partial + \lambda)c, \quad [G_\lambda G] = (\partial + 2\lambda)G - \frac{1}{6}\lambda^3 \mathbf{1}.$$

Thus G has central charge -2 and the stated weights. The affine and ghost factors commute. The mixed terms between S and $\frac{1}{2}\partial h$ supply $(\partial + 2\lambda)\frac{1}{2}\partial h$, and

$$\left[\frac{1}{2}\partial h_\lambda \frac{1}{2}\partial h \right] = -\frac{k}{2}\lambda^3 \mathbf{1}.$$

Consequently

$$[T_\lambda T] = (\partial + 2\lambda)T + \frac{C(k)}{12}\lambda^3 \mathbf{1}. \quad (1.32)$$

The zero mode of ∂h vanishes, so $T_{(0)} = \partial$. The weight statements follow from

$$\begin{aligned} [T_\lambda e] &= \partial e, & [T_\lambda h] &= (\partial + \lambda)h - k\lambda^2 \mathbf{1}, & [T_\lambda f] &= (\partial + 2\lambda)f, \\ [T_\lambda b] &= \partial b, & [T_\lambda c] &= (\partial + \lambda)c. \end{aligned}$$

The extra term in the h bracket is a third-order pole; weight one does not make h primary.

Directly from (1.29),

$$DS =: e\partial c :, \quad D\left(\frac{1}{2}\partial h\right) = -\partial : ec :, \quad DG =: (\partial e)c :.$$

Their sum is zero. Since $e - \mathbf{1}$ is primary of weight zero, c is primary of weight one, and their mutual bracket vanishes, the Wick identity gives $[T_\lambda q] = (\partial + \lambda)q$. Equivalently, the conformal modes commute with its residue D . $\square$

The affine ghosts in (1.29) have weights $(0, 1)$. The reparametrization ghosts of weights $(2, -1)$ define a different complex. In particular, the identity $D^2 = 0$ above imposes no equation $C(k) = 26$.

1.18 A REDUCED COMPLEX AND ITS COHOMOLOGY

Put

$$j =: bc :, \quad H = h + 2j, \quad F = f, \quad E = e - \mathbf{1}, \quad a = \frac{t-1}{2}.$$

The elementary ghost identities needed below are

$$[j_\lambda j] = \lambda \mathbf{1}, \quad [j_\lambda c] = -c, \quad : j c := -\partial c, \quad : j j := (\partial b)c : - : b \partial c :. \quad (1.33)$$

They follow by one or two contractions in the defining Clifford algebra. Let C_k^- be the vertex subalgebra generated by H, F, c , and let C_k^+ be the vertex subalgebra generated by E, b . Their relevant brackets and differentials are

$$\begin{aligned} [H_\lambda H] &= 2t\lambda \mathbf{1}, & [H_\lambda F] &= -2F, & [H_\lambda c] &= -2c, \\ DH &= -2c, & DF &=: Hc : + t\partial c, & Dc &= 0, \\ DE &= 0, & Db &= E. \end{aligned} \quad (1.34)$$

All other brackets among F, c vanish; all brackets in C_k^+ vanish. Both subalgebras are subcomplexes. Define

$$W = F + \frac{1}{4} : HH : + a\partial H, \quad L = t^{-1}W.$$

LEMMA 1.18.1. The state W is closed. The state L satisfies the Virasoro bracket of central charge $C(k)$, and

$$T - L = t^{-1}D(: bf :). \quad (1.35)$$

Proof. Quasi-commutativity gives $cH =: Hc : + 2\partial c$. Therefore

$$D\left(\frac{1}{4} : HH :\right) = - : Hc : - \partial c, \quad D(a\partial H) = -2a\partial c.$$

Equation (1.34) and $2a = t - 1$ prove $DW = 0$.

For the Virasoro bracket, set $P = \frac{1}{4} : HH : + a\partial H$. The Heisenberg calculation using $[H_\lambda H] = 2t\lambda$ gives

$$[P_\lambda P] = t(\partial + 2\lambda)P + \left(\frac{t^2}{12} - 2ta^2\right)\lambda^3 \mathbf{1}.$$

The Wick identity and $[F_\lambda H] = 2F$ give

$$[F_\lambda P] = : HF : + t(\partial + \lambda)F, \quad [P_\lambda F] = - : HF : + t\lambda F.$$

Since $[F_\lambda F] = 0$, division by t^2 yields

$$[L_\lambda L] = (\partial + 2\lambda)L + \frac{1 - 24a^2/t}{12}\lambda^3 \mathbf{1}.$$

Here $1 - 24a^2/t = 13 - 6(t + t^{-1})$.

Finally, $: fe := ef : - \partial h$ and (1.33) give

$$\begin{aligned} t(T - L) &= (e - \mathbf{1})f : - : hbc : - k : b\partial c : \\ &= D(: bf :). \end{aligned}$$

All products in the middle expression use the affine-ghost tensor factors, so moving h or f through a ghost introduces no contraction. $\square$

THEOREM 1.18.2. For every $k \neq -2$, there is an injective quasi-isomorphism of unital differential vertex superalgebras

$$\iota_T : (\text{Vir}^{C(k)}, 0) \longrightarrow (C_k, D), \quad \iota_T(\omega) = T, \quad (1.36)$$

where $\text{Vir}^{C(k)}$ is the universal Virasoro vacuum vertex algebra and ω its conformal generator. The map is conformal for (1.31). In particular,

$$H^r(C_k, D) = 0 \quad (r \neq 0), \quad H^0(C_k, D) \simeq \text{Vir}^{C(k)}.$$

Proof. The bracket (1.32) and $DT = 0$ construct the unital map (1.36) by the universal Virasoro presentation.

Normally ordered multiplication defines a linear map of complexes

$$C_k^- \otimes C_k^+ \longrightarrow C_k, \quad u \otimes v \longmapsto u_{(-1)}v. \quad (1.37)$$

This map is a linear isomorphism by the following PBW argument. Give derivatives of e, b, h, c, f filtration degrees 0, 0, 1, 1, 2, respectively. The affine and ghost brackets lower the sum of these degrees by at least one. The associated graded algebra is therefore the polynomial algebra in the even jets and the exterior algebra in the odd jets. The change of generators

$$(e, h, f, b, c) \longmapsto (E, H, F, b, c) = (e - \mathbf{1}, h + 2bc, f, b, c)$$

and all its derivatives is an invertible change in this commutative algebra. Thus the associated graded of (1.37) is an isomorphism. Induction on the nonnegative filtration degree proves the assertion on the original vector spaces. Each state has finite filtration degree. Equation (1.30) proves that the map is a map of complexes, using the tensor sign $(-1)^{|u|}$ on the second differential. It is not asserted to be a tensor-product decomposition of vertex algebras.

Write $E_r = \partial^r E$ and $b_r = \partial^r b$ for $r \geq 0$ in the commutative vertex algebra C_k^+ . Then

$$C_k^+ = \mathbb{C}[E_r : r \geq 0] \otimes \Lambda(b_r : r \geq 0), \quad D = \sum_{r \geq 0} E_r \frac{\partial}{\partial b_r}.$$

The odd operator $s = \sum_{r \geq 0} b_r \partial / \partial E_r$ satisfies $Ds + sD = N$, where N counts the total number of E_r, b_r factors. Every polynomial contains finitely many factors and variables. On its positive- N subspace, s/N is a contraction. Thus $\mathbb{C}\mathbf{1} \rightarrow C_k^+$ is a quasi-isomorphism and the inclusion $C_k^- \rightarrow C_k$ is a quasi-isomorphism by (1.37).

For C_k^- , use the nonnegative filtration in which each derivative of H, c has degree one and each derivative of F degree two. Its associated graded differential is

$$dH_r = -2c_r, \quad dF_r = \partial^r (H_0 c_0), \quad dc_r = 0.$$

Introduce even variables $w_r = \partial^r (F_0 + H_0^2/4)$. The map $(F_r, H_r, c_r) \mapsto (w_r, H_r, c_r)$ is polynomial and invertible, and $dw_r = 0$. Consequently the associated graded complex is

$$\mathbb{C}[w_r : r \geq 0] \otimes \left(\mathbb{C}[H_r : r \geq 0] \otimes \Lambda(c_r : r \geq 0), -2 \sum_{r \geq 0} c_r \frac{\partial}{\partial H_r} \right). \quad (1.38)$$

On the second factor, $s_- = -\frac{1}{2} \sum_r H_r \partial / \partial c_r$ gives $ds_- + s_- d = N_-$, the number of H_r, c_r factors. This proves that its cohomology consists of constants.

The Virasoro relation for L constructs a map $\iota_L : \text{Vir}^{C(k)} \rightarrow C_k^-$. Filter its source by assigning degree two to all derivatives of ω . The PBW basis of the universal Virasoro vacuum module consists of the ordered products of $L_{-n}, n \geq 2$, acting on the vacuum. Its associated graded algebra is $\mathbb{C}[\partial^r \omega : r \geq 0]$. The associated graded of ι_L sends $\partial^r \omega$ to w_r/t and is a quasi-isomorphism by (1.38). The associated graded of its mapping cone is acyclic. For a cone cycle, solve its boundary equation in its highest filtration quotient and subtract that boundary. Induction terminates because the filtration is bounded below. Hence ι_L is a quasi-isomorphism.

Equation (1.35) identifies $[T] = [L]$ in $H^0(C_k)$. The two induced vertex homomorphisms agree on the universal generator, hence agree everywhere. Thus ι_T is a quasi-isomorphism as well. Since its source has zero differential, an element in its kernel has zero cohomology class in the source and must be zero. This proves injectivity. $\square$

Theorem 1.18.2 concerns the universal vacuum carrier. It agrees with the PBW and reduction statements in [3, Theorem 5.8, p. 32; §5.3, p. 33; Example 5.22, p. 39]. No vanishing statement on an unrestricted category of affine modules is used in the proof. The sign of the character in (1.29) differs from the $e + 1$ convention of [3, §5.1]; the substitution $(e, f, b, c) \mapsto (-e, -f, -b, -c)$ identifies the two conventions.

PROPOSITION 1.18.3 (*Direction of a strict comparison*). There is no unital vertex superalgebra homomorphism from C_k to a purely even vertex algebra. In particular, ι_T has no strict unital vertex retraction $C_k \rightarrow \text{Vir}^{C(k)}$.

Proof. Parity preservation sends b, c to zero. The identity $b_{(0)}c = \mathbf{1}$ would then send $\mathbf{1}$ to zero, contradicting the unit condition. $\square$

The inverse to (1.36) exists after inverting quasi-isomorphisms. The ordinary projection from closed states to $H^0(C_k)$ is defined on $\ker D$, not on every state of C_k as a vertex homomorphism.

1.19 THE UNITAL COLLISION MAP ON AN AFFINE COORDINATE CHART

Let $X = \mathbb{A}_{\mathbb{C}}^1$ with coordinate z . For a differential vertex superalgebra (V, d) , define a left D_X -module

$$\mathcal{M}_V = \mathcal{O}_X \otimes_{\mathbb{C}} V, \quad \nabla_{\partial_z}(f \otimes v) = f' \otimes v + f \otimes \partial v,$$

with internal differential $\mathbf{1} \otimes d$. Its corresponding right module is $\mathcal{A}_V = \omega_X \otimes \mathcal{M}_V$; explicitly,

$$(f dz \otimes v) \cdot \partial_z = -f' dz \otimes v - f dz \otimes \partial v. \quad (1.39)$$

These modules use algebraic tensor products.

On X^2 , use coordinates z, w and put $x = z - w$. Let $j_2 : X^2 \setminus \Delta \hookrightarrow X^2$ and $\Delta : X \hookrightarrow X^2$ be the open and diagonal embeddings. In left-module notation, set

$$\mathcal{P}_V = \mathcal{O}_{X^2}(*\Delta) \otimes V \otimes V, \quad \mathcal{R}_V = V \otimes \mathbb{C}[w] \otimes x^{-1}\mathbb{C}[x^{-1}].$$

The second expression is the coordinate presentation of the module supported on the diagonal, with actions

$$\partial_z = \partial_x, \quad \partial_w = \partial_w^{\text{base}} - \partial_x + \partial_V. \quad (1.40)$$

Multiplication by x is interpreted modulo regular terms. Tensoring these left modules with $dz \wedge dw$ identifies their right-module versions with $j_{2*}j_2^*(\mathcal{A}_V \boxtimes \mathcal{A}_V)$ and $\Delta_!\mathcal{A}_V$. The residue orientation is $dz \wedge dw = dx \wedge dw$ and $\text{Res } x^{-1}dx = \mathbf{1}$.

Define the full binary collision morphism by

$$\mu_V(f(z, w)a \boxtimes d) = \text{PP}_x(f(w + x, w)Y(a, x)d). \quad (1.41)$$

Here PP_x discards nonnegative powers of x . For fixed a, d and a meromorphic coefficient f of finite pole order, only finitely many modes contribute to the principal part. In particular, this formula also uses negative vertex products when f has a pole. It keeps the vacuum terms of all nonnegative products.

LEMMA 1.19.1. Formula (1.41) is a morphism of differential D_{X^2} -modules. Its right-module form is the unital chiral product on $\mathcal{A}_V$.

Proof. For ∂_z , the chain rule and $Y(\partial a, x) = \partial_x Y(a, x)$ identify the source connection with the first action in (1.40). For ∂_w , use

$$\partial_V Y(a, x)d - \partial_x Y(a, x)d = Y(a, x)\partial_V d.$$

This gives the second action in (1.40). Taking principal parts preserves both equations. The differential identity follows from (1.30), with D replaced by d . Skew symmetry and the three-input Jacobi identity are the corresponding vertex identities after passage to principal parts; right-module conversion supplies the ordinary antisymmetry sign, together with the parity sign. The vacuum embedding is $\omega_X \rightarrow \mathcal{A}_V$, $f dz \mapsto f dz \otimes \mathbf{1}$. For example,

$$\text{PP}_x(x^{-1}Y(\mathbf{1}, x)a) = x^{-1}a,$$

which is the unit collision under the residue identification. This construction is the coordinate expression of [4, §7, Theorem 7.1 and Corollary 7.2, pp. 875–34–35]. $\square$

THEOREM 1.19.2 (*Chiral comparison on an affine chart*). For $k \neq -2$, formula

$$\mathcal{I}_X : \mathcal{A}_{\text{Vir}^{C(k)}} \longrightarrow \mathcal{A}_{C_k}, \quad f dz \otimes v \longmapsto f dz \otimes \iota_T(v)$$

defines a unital quasi-isomorphism of differential chiral algebras on $\mathbb{A}^1$. It preserves the full binary collision morphism:

$$(\Delta! \mathcal{I}_X) \mu_{\text{Vir}} = \mu_{C_k} j_{2*} j_2^* (\mathcal{I}_X \boxtimes \mathcal{I}_X). \quad (1.42)$$

Every finite iterated collision, with its meromorphic coefficient and its vacuum outputs, satisfies the corresponding intertwining identity.

Proof. The state map ι_T commutes with translation, the vacuum, and the internal differentials. It therefore respects (1.39) and the unit embedding. For every $n \in \mathbb{Z}$,

$$\iota_T(a_{(n)}d) = \iota_T(a)_{(n)}\iota_T(d).$$

Substitution into (1.41) proves (1.42) for every coefficient f , including all its regular and polar parts. Induction on the number of binary operations proves the iterated identities. The underlying sheaf map is a quasi-isomorphism because Theorem 1.18.2 is a quasi-isomorphism of complexes of vector spaces and tensoring with each $\mathcal{O}_X$ -stalk is exact. $\square$

The lowest Virasoro collision reads

$$Y(T, x)T \sim \frac{C(k)}{2x^4} \mathbf{1} + \frac{2T}{x^2} + \frac{\partial T}{x}.$$

Thus $T_{(3)}T = C(k)\mathbf{1}/2$, $T_{(2)}T = 0$, $T_{(1)}T = 2T$, and $T_{(0)}T = \partial T$. For a regular coefficient $g(x)$, the residue is

$$\text{Res}_{x=0} g(x)Y(T, x)T dx = \frac{C(k)}{12} g'''(0) \mathbf{1} + 2g'(0)T + g(0)\partial T.$$

At $k = -8/3$ or $-7/2$, taking $g = x^3$ gives $13\mathbf{1}$. The independent ghost collision is $\text{Res } Y(b, x)c dx = \mathbf{1}$. Neither vacuum output can be removed by a linear projection while preserving the original product.

1.20 THE BINARY COLLISION COMPLEX

For any V above, put

$$P_V = j_{2*} j_2^* (\mathcal{A}_V \boxtimes \mathcal{A}_V), \quad R_V = \Delta! \mathcal{A}_V.$$

Their internal differentials are $d_P(a \boxtimes b) = da \boxtimes b + (-1)^{|a|} a \boxtimes db$ and d_R induced by d . Define the unital binary collision complex

$$K_2(V) = P_V[1] \oplus R_V, \quad d_{K_2}(sp, r) = (-sd_P p, \mu_V(p) + d_R r). \quad (1.43)$$

The suspension convention is $|sp| = |p| - 1$. The equality $d_R \mu_V = \mu_V d_P$ proves $d_{K_2}^2 = 0$.

COROLLARY 1.20.1. The componentwise map

$$K_2(\text{Vir}^{C(k)}) \longrightarrow K_2(C_k)$$

defined by $\mathcal{I}_X \boxtimes \mathcal{I}_X$ on $P[1]$ and $\Delta! \mathcal{I}_X$ on R is a quasi-isomorphism.

Proof. Equation (1.42) proves the total chain identity, including the off-diagonal component μ . Tensor products of complexes over $\mathbb{C}$ preserve quasi-isomorphisms. The localization $\mathcal{O}_{X^2} \rightarrow \mathcal{O}_{X^2}(*\Delta)$ is flat, so the map on P is a quasi-isomorphism. The coordinate presentation (1.40) identifies the map on R with a direct sum of copies of the state map tensored with $\mathcal{O}_X$, so it is a quasi-isomorphism as well. The cone of the total map has a two-step filtration with these two acyclic cones as quotients. It is acyclic by the long exact sequence of cohomology. $\square$

The complex (1.43) retains the unit and is explicitly defined on X^2 . Its comparison does not require an augmentation of C_k . An identification with a reduced chiral bar defined using an augmentation kernel would require such an augmentation, which is excluded by $b_{(\circ)}c = \mathbf{1}$. An infinite totalization over configuration spaces requires its own definition and convergence theorem; the two-step argument above contains no such limit.

1.21 DESCENT FOR THE DRINFELD–SOKOLOV CONFORMAL STRUCTURE

The conformal structure (1.31) also specifies coordinate changes. Let X now be a smooth complex algebraic curve, and let $\widehat{X} \rightarrow X$ be its torsor of formal coordinates. The Lie algebra of $\text{Aut } \mathbb{C}[[z]]$ acts on C_k by

$$-z^{n+1}\partial_z \mapsto L_n \quad (n \geq 0). \quad (1.44)$$

The Virasoro central term vanishes on this subalgebra. All PBW states have nonnegative integral L_0 -weights, and L_n lowers weight by $n > 0$. The orbit of a weight- N state lies in the span of the finitely many words in $L_1, \dots, L_N$ with total weight loss at most N . The positive part therefore integrates to formal coordinate changes, and integral L_0 integrates to $\mathbb{G}_m$. The action is locally finite despite the infinite-dimensional weight-zero subspace.

Form the associated sheaf

$$\mathcal{M}_{C_k, X}^{\text{DS}} = \widehat{X} \times_{\text{Aut } \mathbb{C}[[z]]} C_k, \quad \mathcal{A}_{C_k, X}^{\text{DS}} = \omega_X \otimes \mathcal{M}_{C_k, X}^{\text{DS}}.$$

Here $(ug, v) \sim (u, gv)$ for the right coordinate torsor and the left action (1.44). The translation operator extends (1.44) to regular formal vector fields and gives the connection $d + \partial dz$ in a coordinate trivialization. This is the associated-bundle connection described in [4, §4.2, p. 875–18]. The sheaf is the union of its finite-rank associated subbundles as an $\mathcal{O}_X$ -module; the connection can raise their filtration.

PROPOSITION 1.21.1. For every smooth complex algebraic curve X and $k \neq -2$, the map ι_T descends to a unital quasi-isomorphism of differential chiral algebras

$$\mathcal{A}_{\text{Vir}^{C(k)}, X} \longrightarrow \mathcal{A}_{C_k, X}^{\text{DS}}.$$

The local binary collision comparison of Corollary 1.20.1 descends to X^2 .

Proof. The conformal map ι_T intertwines all L_n and hence their integrated coordinate actions. Proposition 1.17.1 says that D commutes with the same action. Thus both the sheaf map and its differential are defined on the associated sheaves. Translation compatibility gives horizontality. The coordinate covariance of the vertex operation, equivalently the chiral construction in Lemma 1.19.1, glues the collision identity on overlaps. Its unit is preserved before descent. Locally in an étale coordinate, the underlying complex is $\mathcal{O}_X \otimes C_k$, so Theorem 1.18.2 proves the quasi-isomorphism. The same local argument applies to the two terms of (1.43); quasi-isomorphism is local. $\square$

The superscript DS specifies the coordinate action, which uses T . For example, e and b have weight zero, f has weight two, and h has third-order conformal pole $-2k\mathbf{1}$. These are not the three weight-one primary currents of the Sugawara coordinate action. The proposition therefore does not identify this associated sheaf with a separately prescribed affine-current bundle using other coordinate transitions. Nor does it construct compatibility with clutching at nodal curves, a completed chiral bar, or a second semi-infinite differential. Each such comparison has an additional carrier and an additional intertwining equation.

1.22 UNITAL VIRASORO CHIRAL CHAINS ON THE PROJECTIVE LINE

The vacuum term in a collision contributes to chiral chains on the projective line. For the critical Virasoro algebra with ghosts, its differential admits a simpler description through global vector fields. The resulting cohomology has generators in degrees one and three. Both the collision differential and the degree shifts enter this calculation.

1.23 THE CRITICAL DIFFERENTIAL AND THE GHOST SUBALGEBRA

Work over $\mathbb{C}$, with algebraic tensor products and direct sums. Set $X = \mathbb{P}^1$. Complexes are cohomological, with $K[r]^j = K^{j+r}$ and differential $(-1)^r d_K$. The sign for interchanging homogeneous elements of degrees a, b is $(-1)^{ab}$. Ghost degree gives both cohomological degree and parity modulo two. Conformal weight is a separate integer grading.

Let $V = \text{Vir}_{26}$ be the universal Virasoro vacuum vertex algebra. Its vacuum $\mathbf{1}$ satisfies $L_n \mathbf{1} = 0$ for $n \geq -1$. The ordered monomials in L_{-n} , $n \geq 2$, form its vector-space basis. Write $T = L_{-2}\mathbf{1}$. Let $\mathcal{F}$ be the bc vertex superalgebra with

$$\{b_m, c_n\} = \delta_{m+n,0}, \quad \{b_m, b_n\} = \{c_m, c_n\} = 0, \quad b_n \Omega = 0 \ (n \geq -1), \quad c_n \Omega = 0 \ (n \geq 2).$$

Its creators are b_n for $n \leq -2$ and c_n for $n \leq 1$. The generating states are $b = b_{-2}\Omega$ and $c = c_1\Omega$, of degrees $-1, 1$ and weights $2, -1$. Put $A = V \otimes \mathcal{F}$ and write Ω also for its vacuum. The field conventions are

$$\begin{aligned} Y(T, z) &= \sum_n L_n z^{-n-2}, & Y(b, z) &= \sum_n b_n z^{-n-2}, \\ Y(c, z) &= \sum_n c_n z^{1-n}, & Y(a, z) &= \sum_n a_{(n)} z^{-n-1}. \end{aligned}$$

In particular, the vertex product $b_{(0)}$ is the conformal mode b_{-1} and differs from the constraint operator b_0 .

Normal ordering moves all creators to the left with their odd permutation signs and omits contractions. On A , define

$$Q = \sum_n c_{-n} L_n - \frac{1}{2} \sum_{m,n} (m-n) : c_{-m} c_{-n} b_{m+n} :.$$

These sums are locally finite on each algebraic state. The total stress tensor and its zero mode are

$$T^{\text{tot}} = T - 2 : b \partial c : - : \partial b c :, \quad H = L_0 + L_0^{\text{gh}}.$$

Write $Y(T^{\text{tot}}, z) = \sum_n \mathcal{L}_n z^{-n-2}$, so $H = \mathcal{L}_0$. Theorem 1.3.2 gives $Q^2 = 0$ and $\{Q, b_0\} = H$. The field in (1.15) has zero mode Q . The vertex commutator identity for a zero mode therefore gives

$$Q(a_{(n)} a') = (Qa)_{(n)} a' + (-1)^{|a|} a_{(n)} Qa'. \quad (1.45)$$

In particular, Q commutes with translation ∂ , and

$$Qc =: c\partial c :, \quad Qb = T^{\text{tot}}, \quad QT =: c\partial T : + 2 : \partial c T : + \frac{13}{6} \partial^3 c. \quad (1.46)$$

For the last identity, take the residue of $c(z)T(z)T(w)$. The coefficients of $(z-w)^{-4}$, $(z-w)^{-2}$, $(z-w)^{-1}$ are 13 , $2T$, ∂T . Taylor expansion of $c(z)$ gives the stated coefficients.

Let $B \subset A$ be the subalgebra generated by c and its derivatives. It is the commutative vertex algebra

$$B = \Lambda(c^{(j)} : j \geq 0), \quad c^{(j)} = \partial^j c = j! c_{1-j} \Omega, \quad Qc^{(j)} = \partial^j (c\partial c). \quad (1.47)$$

Every generator has degree one. All singular products between these generators vanish. The regular vertex operation is $Y(f, t)g = (e^{t\partial} f)g$.

The total central charge is zero. The identities $[Q, \mathcal{L}_n] = 0$ in (1.14) permit coordinate descent of (A, Q) . The c -ghosts are preserved by these coordinate changes. For a change $w = w(z)$ their field transforms by

$$c_w(w) = \frac{dw}{dz} c_z(z). \quad (1.48)$$

Repeated differentiation determines every jet transition in B . Denote the resulting left state modules by $\mathcal{A}^\ell, \mathcal{B}^\ell$. The second is a commutative differential graded $\mathcal{D}_X$ -algebra. Their right modules are $\mathcal{A} = \omega_X \otimes \mathcal{A}^\ell$ and $\mathcal{B} = \omega_X \otimes \mathcal{B}^\ell$. They carry the usual chiral products associated with their vertex algebras. In a coordinate, the connection on a state section is $\partial_z + \partial$. Thus the line generated by c in the left module is ω_X . The connection and the primary-state line convention agree with Frenkel [6], §§4.2–4.3, pp. 875–18–875–19. The conformal weights here are bounded below by -1 . Positive Virasoro modes lower weight and hence act locally nilpotently, so the same coordinate-bundle construction applies.

PROPOSITION 1.23.1. The inclusion $\mathcal{B} \rightarrow \mathcal{A}$ is a quasi-isomorphism of unital differential graded chiral superalgebras.

Proof. The formulas above prove compatibility with the unit, products, differential, and coordinate changes. It remains to calculate the cohomology in a coordinate trivialization. Both A and B have integral energy decompositions with finite energy support on each element. On a nonzero energy λ , the operator $\lambda^{-1}b_0$ contracts both complexes. On B , b_0 differentiates the exterior variable c_0 and preserves B . The contraction identity is the restriction of $\{Q, b_0\} = H$.

The matter algebra has $V_0 = \mathbb{C}\mathbf{1}$, $V_1 = 0$, and no negative weights. The only negative-energy ghost creator is c_1 , which occurs at most once. Every b creator has energy at least two. Consequently the energy-zero subspaces of A and B are identical, with basis

$$e = \Omega, \quad u = c_0 \Omega, \quad v = c_{-1} c_1 \Omega, \quad w = c_{-1} c_0 c_1 \Omega.$$

Their degrees are 0, 1, 2, 3, and the differential is

$$Qe = 0, \quad Qu = -2v, \quad Qv = Qw = 0. \quad (1.49)$$

Here Qu follows from $\{Q, c_0\} = -2 \sum_{n>0} n c_{-n} c_n$. Nilpotence gives $Qv = 0$, and degree gives $Qw = 0$. The map $h(v) = -u/2$, zero on the other three basis vectors, contracts the pair u, v . Thus both cohomologies have basis $[e]$, $[w]$, and inclusion identifies them. This local calculation proves the assertion for the underlying sheaf complexes. The contraction need not be invariant under coordinate changes for this purpose. $\square$

1.24 GLOBAL VECTOR FIELDS AND A CONSTANT DIFFERENTIAL GRADED ALGEBRA

Let $\mathfrak{g} = H^0(X, T_X)$ with its vector-field bracket. On the affine chart with coordinate z , use

$$e_- = \partial_z, \quad e_0 = z\partial_z, \quad e_+ = z^2\partial_z.$$

Then $[e_-, e_0] = e_-$, $[e_-, e_+] = 2e_0$, and $[e_0, e_+] = e_+$. Let $E = C_{\text{CE}}^\bullet(\mathfrak{g}, \mathbb{C})$ be its Chevalley–Eilenberg cochain algebra. If α, β, γ are the dual degree-one generators, its differential is

$$E = \Lambda(\alpha, \beta, \gamma), \quad d\alpha = -\alpha\beta, \quad d\beta = -2\alpha\gamma, \quad d\gamma = -\beta\gamma. \quad (1.50)$$

Write $E_X^\ell = \mathcal{O}_X \otimes E$ with the connection on the first factor, and $E_X = \omega_X \otimes E$.

PROPOSITION 1.24.1. There is a quasi-isomorphism of unital differential graded chiral algebras

$$\epsilon : \mathcal{B} \longrightarrow E_X.$$

In the coordinate z , it is determined on left modules by

$$\epsilon(c) = -f(z), \quad f(z) = \alpha + z\beta + z^2\gamma, \quad \epsilon(\partial^j c) = \partial_z^j \epsilon(c). \quad (1.51)$$

Proof. The last formula is the equation $\partial_z \epsilon(c) = \epsilon(\nabla_{\partial_z} c)$ for a morphism of left $\mathcal{D}_X$ -modules. Thus $\epsilon(\partial c) = -f'(z)$ and $\epsilon(\partial^2 c) = -2\gamma$. Equation (1.50) gives $df = -ff'$. Consequently

$$d\epsilon(c) = ff', \quad \epsilon(Qc) = (-f)(-f') = ff'.$$

The two differentials commute with translation, so they agree on every jet. Both are derivations, so they agree on B .

The vector field $f(z)\partial_z$ is the tautological element of $\mathfrak{g} \otimes \mathfrak{g}^*$, with its coefficients placed in degree one. Under $w = w(z)$, its coefficient becomes $f_w = w'f_z$. This agrees with (1.48), and the jet equations agree by differentiation. For the two standard charts $w = 1/z$, one obtains

$$\epsilon(c_w) = \alpha w^2 + \beta w + \gamma = (-z^{-2})\epsilon(c_z).$$

The formulas are regular in both charts. They therefore define the claimed global morphism. Commutative vertex products are determined by multiplication and translation, so this is also a chiral-algebra morphism.

The differential $E^1 \rightarrow E^2$ in (1.50) is an isomorphism. Its matrix in the ordered bases (α, β, γ) and $(\alpha\beta, \alpha\gamma, \beta\gamma)$ is $\text{diag}(-1, -2, -1)$. The other nonzero cohomology classes of E are 1 and $\eta = \alpha\beta\gamma$. At $z = 0$,

$$\epsilon(w) = \epsilon(\tfrac{1}{2}(\partial^2 c)(\partial c)) = (-\gamma)(-\beta)(-\alpha) = \eta.$$

The same product equals η at every finite z . In the w chart its value is also η . The four-state calculation in (1.49) therefore proves that ϵ induces an isomorphism on every cohomology stalk. $\square$

Let $K = \Lambda(\eta)$ with $|\eta| = 3$ and zero differential. The map $K \rightarrow E$ given by $\eta \mapsto \alpha\beta\gamma$ is a unital quasi-isomorphism of commutative differential graded algebras. Combining the two propositions gives a zigzag of actual chiral-algebra morphisms

$$\mathcal{A} \longleftarrow \mathcal{B} \xrightarrow{\epsilon} E_X \longleftarrow K_X, \quad K_X = \omega_X \otimes K. \quad (1.52)$$

Every arrow is a quasi-isomorphism. The arrows have their displayed directions: a chosen inverse exists only after passing to the derived category.

1.25 THE CHEVALLEY–COUSIN DIFFERENTIAL

For a nonempty finite set I , let $\text{Part}(I)$ be its set of partitions. Regard a partition π as the quotient map from I to its set of blocks. Let $\Delta_\pi : X^\pi \hookrightarrow X^I$ identify coordinates within each block. Let $U^\pi \subset X^\pi$ be the open set of distinct block coordinates, with inclusion j_π . For a right differential graded chiral algebra $\mathcal{M}$, define

$$C(\mathcal{M})_{X^I} = \bigoplus_{\pi \in \text{Part}(I)} (\Delta_\pi)_* (j_\pi)_* j_\pi^* (\mathcal{M}[1])^{\boxtimes \pi}. \quad (1.53)$$

Here Δ_* is direct image for right $\mathcal{D}$ -modules. The open direct image allows poles of arbitrary finite order along the omitted diagonals. Every summand is algebraic: no inverse limit in pole order or arity is taken. Formula (1.53) is Beilinson–Drinfeld’s complex [5], equation (3.4.11.1), p. 182.

Write sa for an element of $\mathcal{M}[1]$, so $|sa| = |a| - 1$. The internal differential is the tensor differential extending $sa \mapsto -s(d_{\mathcal{M}}a)$. The collision differential merges two blocks using the suspended chiral product

$$d_2(sa\,sb) = (-1)^{|a|} s\mu(a, b). \quad (1.54)$$

For any pair of blocks, move their suspended factors to the first two positions using the Koszul rule, apply (1.54), and retain the other factors. The sum over all pairs defines d_{col} . In (1.54), the argument is localized away from the two-block diagonal and the output is its direct image. Localizations at the other diagonals remain in place. This specifies a morphism of right $\mathcal{D}_{X^I}$ -modules on each summand.

PROPOSITION 1.25.1. With $d = d_{\text{int}} + d_{\text{col}}$, formula (1.53) defines a complex. The diagonal structure maps and the factorization identifications commute with d . The construction is functorial for differential graded chiral-algebra morphisms.

Proof. The internal square is zero by the differential on $\mathcal{M}$. Equation (1.45), with the suspension signs in (1.54), gives $d_{\text{int}}d_{\text{col}} + d_{\text{col}}d_{\text{int}} = 0$. Two mergers of disjoint pairs cancel in opposite orders because each merger has degree one. For three blocks, the three terms in the remaining square are the suspended chiral Jacobi identity. These cases exhaust all terms in d_{col}^2 .

A diagonal structure map includes the summands whose partitions factor through the specified quotient of labels. Merging blocks commutes with this inclusion. Over separated groups of points, partitions split into the partitions of those groups. The tensor-product identification of the corresponding summands commutes with internal differentials and collisions inside each group. For three groups, both composite identifications use the same partition and the same tensor symmetry. They therefore agree. The same argument with a morphism on every block proves functoriality. $\square$

For $|I| = 2$, the complex contains precisely

$$j_* j^* (\mathcal{M}[1] \boxtimes \mathcal{M}[1]) \xrightarrow{d_{\text{col}}} \Delta_* \mathcal{M}[1].$$

For $|I| = 3$, its terms have three, two, and one blocks. The two-block term has one summand for each pair of labels. Thus the first coherence after a binary collision is the three-block Jacobi identity, with all three intermediate diagonals present.

1.26 AN ACTUAL GLOBAL CHAIN COMPLEX

The de Rham functor on right $\mathcal{D}$ -modules has the convention

$$\text{DR}_X(\mathcal{M}) = [\mathcal{M} \otimes_{\mathcal{O}_X} T_X \longrightarrow \mathcal{M}]$$

in auxiliary degrees $-1, 0$. In particular, $\mathrm{DR}_X(\omega_X) = \Omega_X^\bullet[1]$. To keep collision operations strict after taking sections, choose the following two resolutions.

Let $P^\circ = \mathrm{Sym}_{\mathcal{O}_X}^{\geq 0}(\mathcal{D}_X)$, where the left $\mathcal{D}_X$ -action is the diagonal action. Its augmentation sends a differential operator D in the generating summand to $D(1)$. Set

$$P^{-1} = \ker(P^\circ \rightarrow \mathcal{O}_X), \quad d_P : P^{-1} \hookrightarrow P^\circ.$$

Multiplication in degree zero is the symmetric-algebra multiplication. Its action on P^{-1} is the ideal action, and the product of two degree-minus-one elements is zero. The Leibniz identity on two such elements is $uv - uv = 0$. This defines a nonunital commutative differential graded $\mathcal{D}_X$ -algebra resolution of $\mathcal{O}_X$. Its terms are $\mathcal{D}_X$ -flat by the curve construction in [5], §2.2.10, p. 72. Nonunitality belongs to this auxiliary resolution and imposes no augmentation on $\mathcal{A}$.

Use the two affine charts U_o, U_∞ and their intersection U_{oi} . Write j_i, j_{oi} for their embeddings in X . Let $\mathcal{Q}^\circ$ consist of triples

$$(f_o, f_\infty, F(s)) \in j_{o*}\mathcal{O}_{U_o} \oplus j_{\infty*}\mathcal{O}_{U_\infty} \oplus j_{oi*}\mathcal{O}_{U_{oi}}[s]$$

such that $F(o) = f_o|_{U_{oi}}$ and $F(1) = f_\infty|_{U_{oi}}$. Set $\mathcal{Q}^1 = j_{oi*}\mathcal{O}_{U_{oi}}[s] ds$, with differential $F \mapsto F'(s)ds$. Multiplication is polynomial multiplication with $ds^2 = 0$. The connection differentiates the coefficients, holding s fixed. This is the two-chart Thom–Sullivan resolution from [5], §4.1.3, pp. 278–279. Each term is $\mathcal{O}_X$ -flat. For $\mathcal{Q}^\circ$, this also follows from its exact sequence with quotient $j_{o*}\mathcal{O}_{U_o} \oplus j_{\infty*}\mathcal{O}_{U_\infty}$ and kernel $s(s-1)j_{oi*}\mathcal{O}_{U_{oi}}[s]$.

For any $\mathcal{M}$, put $\mathcal{M}_{PQ} = \mathcal{M} \otimes P \otimes \mathcal{Q}$. The tensor differential on $a \otimes p \otimes q$ is

$$d_{\mathcal{M}}a \otimes p \otimes q + (-1)^{|a|}a \otimes d_P p \otimes q + (-1)^{|a|+|p|}a \otimes p \otimes d_{\mathcal{Q}}q.$$

Tensor symmetry and multiplication in $P, \mathcal{Q}$ extend the chiral product. For a right $\mathcal{D}_Y$ -module N , set $h_Y(N) = N \otimes_{\mathcal{D}_Y} \mathcal{O}_Y$, applied degreewise. Define

$$\mathbb{C}_X(\mathcal{M}) = \mathrm{colim}_I \Gamma(X^I, h_{X^I}(C(\mathcal{M}_{PQ})_{X^I})). \quad (1.55)$$

The index category has nonempty finite sets and arrows opposite to surjections. Its transition maps are the diagonal structure maps. Thus (1.55) is an actual complex of vector spaces with the differential already specified. The comparison of Beilinson–Drinfeld, equation (4.2.12.2), pp. 305–306, identifies its derived-category object with

$$C^{\mathrm{ch}}(X, \mathcal{M}) = R\Gamma_{\mathrm{DR}}(\mathrm{Ran}(X), C(\mathcal{M})).$$

Their comparison is functorial and preserves the filtration by the number of blocks.

As a graded vector space, (1.55) has the smaller description

$$\mathbb{C}_X(\mathcal{M}) = \bigoplus_{n \geq 1} \Gamma(U^{[n]}, h_{X^n}((\mathcal{M}_{PQ}[1])^{\boxtimes n}))_{\mathfrak{S}_n}.$$

This is equation (4.2.12.5), p. 307, with its induced differential. The symmetric group acts on the suspended factors. The subscript means coinvariants over $\mathbb{C}$. The summand with one point contains vacuum states through the unit $\omega_X \rightarrow \mathcal{M}$. There is no arity-zero summand in this reduced Chevalley complex.

1.27 THE FIRST COLLISION AND ITS DIFFERENTIAL

Fix a formal coordinate $t = z_1 - z_2$ transverse to the diagonal. Normalize $\text{Res}(t^{-1}dt) = 1$. For a Laurent polynomial f and states a, a' , the normal residue coefficient of the collision is

$$\rho_f(a, a') = \text{Res}_{t=0} f(t)Y(a, t)a' dt.$$

The whole distribution is its principal part $[f(t)Y(a, t)a']$. This is the local chiral-product formula of [6], Theorem 7.1 and Corollary 7.2, p. 875-35. Pairing this principal part with all powers of t recovers every normal derivative. Lower truncation makes each principal part finite. The vacuum and Virasoro products give

$$\rho_1(b, c) = 1, \quad \rho_{t^3}(T, T) = 131, \quad \rho_{t^{-1}}(1, a) = a. \quad (1.56)$$

All three identities occur in (1.53) and (1.55).

The first mixed differential identity can be computed on b, c without infinite sums:

$$\begin{aligned} Q\rho_1(b, c) &= \rho_1(Qb, c) - \rho_1(b, Qc) \\ &= (T^{\text{tot}})_{(0)}c - b_{(0)}(:c\partial c:) \\ &= \partial c - \partial c = 0. \end{aligned}$$

The second term uses $b_{(0)}c = 1$ and $b_{(0)}\partial c = 0$. The first uses the translation mode of the total stress tensor. For the Virasoro collision,

$$(QT)_{(3)}T = -26\partial c, \quad T_{(3)}QT = 26\partial c.$$

Indeed, differentiating the central term $13t^{-4}$ in the first variable gives $-52t^{-5}$. Its linear Taylor coefficient contributes $-52\partial c$. The term $2\partial c T$ contributes $26\partial c$. In the second expression, only $2\partial c T$ contributes to the coefficient of t^{-4} . This proves $Q\rho_{t^3}(T, T) = 0$ directly.

The general identity

$$Q\rho_f(a, a') = \rho_f(Qa, a') + (-1)^{|a|}\rho_f(a, Qa')$$

follows coefficientwise from (1.45). It also proves binary compatibility for every arrow of (1.52). For ϵ , its compatibility with the regular collision $f = t^{-1}$ reads $\epsilon(aa') = \epsilon(a)\epsilon(a')$. Compatibility with all other f follows from its translation equation. The first three-point compatibility is therefore the commuting diagram obtained by applying each arrow to all three Jacobi composites.

1.28 THE FILTRATION AND ITS CHIRAL SYMBOL

Use the Poincaré–Birkhoff–Witt (PBW) ordering on the negative matter modes and the exterior ghost generators. Give each negative matter mode and each b creator order one. Give every c creator order zero. Let $F_p A$ be the span of ordered monomials of order at most p , with $F_{-1}A = 0$. This is an increasing, exhaustive filtration, with $F_0 A = B$. Translation and coordinate changes preserve it. Normally ordered products add orders, and each singular contraction lowers their sum. Equations (1.46) show that Q preserves F . Its action on an arbitrary ordered monomial follows from (1.45) with $n = -1$.

The associated graded vertex algebra is commutative, with local description

$$R = \mathbb{C}[t^{(j)} : j \geq 0] \otimes \Lambda(b^{(j)}, c^{(j)} : j \geq 0).$$

Here t, b, c are the symbols of T, b, c , their degrees are 0, -1, 1, and their orders are 1, 1, 0. The index (j) means j translations. The differential q commutes with translation and is determined by

$$qc = c\partial c, \quad qb = t - 2b\partial c - (\partial b)c, \quad qt = c\partial t + 2(\partial c)t. \quad (1.57)$$

The term $13\partial^3 c/6$ in QT has lower order and hence disappears from qt . The singular collisions in (1.56) also lower order. Regular collisions remain: for example $\rho_{t^{-1}}(1, a) = a$ on R .

Put $\ell = qb$. The change from t to $\ell = t - 2b\partial c - (\partial b)c$ is an invertible polynomial change of jet generators. Equations (1.57) give $q\ell = 0$ by direct substitution. Thus

$$q = q_B + \sum_{j \geq 0} \ell^{(j)} \frac{\partial}{\partial b^{(j)}}$$

in these generators. Let N count the ℓ and b factors and define the degree-minus-one derivation

$$h = \sum_{j \geq 0} b^{(j)} \frac{\partial}{\partial \ell^{(j)}}.$$

Each sum is finite on a polynomial, and $qh + hq = N$. On the positive-order subspace, $N^{-1}h$ is a contraction. Both b and $\ell = qb$ transform as primary fields of weight two, so this construction respects the jet transition maps. It follows that $\mathcal{B} \hookrightarrow \text{gr}_F \mathcal{A}$ is a quasi-isomorphism, with positive orders acyclic.

Give (1.53) and (1.55) total order $\sum_i p_i$ on a tensor of factors in $F_{p_i} \mathcal{A}$. Keep P, Q in order zero. The symbol operation on each tensor induces the typed isomorphism of actual complexes

$$\sigma : \text{gr}_F C_X(\mathcal{A}) \xrightarrow{\cong} C_X(\text{gr}_F \mathcal{A}), \quad [\boxtimes_i(a_i \otimes p_i \otimes q_i)] \mapsto \boxtimes_i([a_i] \otimes p_i \otimes q_i). \quad (1.58)$$

This is the chain-level comparison in [5], equation (4.2.18.3), p. 311. Its hypotheses hold because the chosen P, Q have flat terms and the state filtration has locally free associated graded pieces. Localization, external tensor product, and diagonal direct image respect these symbols. For the functor h and global sections, the flat-resolution argument of §4.2.12 gives the same comparison. The internal differential becomes (1.57). The collision differential becomes the regular commutative vertex collision on R . These identifications verify that (1.58) commutes with both parts of the differential.

In particular, the first PBW target is the actual complex

$$\text{gr}_0^F C_X(\mathcal{A}) = C_X(\mathcal{B}) = \bigoplus_{n \geq 1} \Gamma(U^{[n]}, h((\mathcal{B}_{PQ}[1])^{\boxtimes n}))_{\mathfrak{S}_n},$$

with (1.47), the P, Q differentials, and the regular collision products specified above. For the separate filtration G by the number of points,

$$\text{gr}_1^G \text{gr}_0^F C_X(\mathcal{A}) = \Gamma(X, h(\mathcal{B}_{PQ}[1])).$$

Thus PBW order and collision length give different filtrations with different first terms.

There is also an actual filtered chain map

$$\iota : C_X(\mathcal{B}) \longrightarrow C_X(\mathcal{A}), \quad (1.59)$$

obtained by including each ghost factor in each partition summand. Give its source the filtration concentrated in order zero. Its associated graded map, followed by σ , is the map induced by $\mathcal{B} \hookrightarrow R$. The contraction above and functoriality of chiral chains show that this associated graded map is a quasi-isomorphism. Thus (1.59) is a filtered quasi-isomorphism. This supplies a lift of the order-zero symbol on the full configuration complex. The differential equation holds on every partition, including all binary and triple collisions.

1.29 THE COHOMOLOGY ON THE PROJECTIVE LINE

THEOREM 1.29.1. For $\mathcal{A}$ associated with $(\text{Vir}_{26} \otimes bc, Q)$ on $X = \mathbb{P}^1$, the algebraic chiral chain complex satisfies

$$C^{\text{ch}}(X, \mathcal{A}) \simeq \Lambda(\xi_1, \xi_3), \quad |\xi_1| = 1, \quad |\xi_3| = 3, \quad (1.60)$$

as an object of the derived category of complexes. The algebra on the right has zero differential. Its cohomology basis is $1, \xi_1, \xi_3, \xi_1\xi_3$, in degrees 0, 1, 3, 4. With Beilinson–Drinfeld’s homological convention $H_a^{\text{ch}} = H^{-a}C^{\text{ch}}$, the corresponding indices are 0, -1 , -3 , -4 . The PBW associated graded chiral complex has the same cohomology, entirely in order zero.

Proof. Apply the functor C_X to (1.52). The result consists of actual chain maps, each a quasi-isomorphism by Beilinson–Drinfeld, §4.2.11 and §4.2.12. It therefore suffices to compute $C^{\text{ch}}(X, K_X)$. The left algebra $K_X^\ell = \mathcal{O}_X \otimes \Lambda(\eta)$ is freely commutative on $\mathcal{O}_X[-3]$. The corresponding right generator module is

$$W = \omega_X[-3].$$

It is $\mathcal{O}_X$ -flat. Set $L = \Gamma(X, h(W_{PQ}))[1]$. The actual comparison is the following map of complexes:

$$\Psi : C_X(\omega_X) \oplus \text{Sym}^{>0}(L) \longrightarrow C_X(K_X). \quad (1.61)$$

On the first summand, use the chain map induced by the chiral unit $\omega_X \rightarrow K_X$. On L , use the inclusion of the generator module into the one-point term. For each positive symmetric degree, take the external product of these inclusions and restrict to the distinct-point locus. The differential commutes with these products because K_X is commutative. This specifies Ψ on every chain, including its entire first summand.

The positive-degree map factors through $C_X(W)$, where W has zero chiral product. Indeed, W is the nonunital free commutative chiral algebra on its odd rank-one generator, and $K_X = W \oplus \omega_X$ is its unitalization. Equation (4.6.2.2) and the proof of the proposition in [5], pp. 342–344, give a quasi-isomorphism $\text{Sym}^{>0}(L) \rightarrow C_X(W)$. The unit and inclusion induce another actual quasi-isomorphism of complexes

$$C_X(\omega_X) \oplus C_X(W) \longrightarrow C_X(K_X)$$

by the proof of [5], §4.4.8, p. 329. Their composition is Ψ , which is therefore a quasi-isomorphism.

The unit contributes through the first complex summand. Its replacement by $\mathbb{C}$ occurs in the derived category, using $C^{\text{ch}}(X, \omega_X) \simeq \mathbb{C}$ from [5], §4.3.3, p. 315. The raw positive-arity chain multiplication has the homotopy-unit structure of §4.3.4. No representative of a cohomology class is required to define (1.61).

The complex L represents

$$R\Gamma_{\text{DR}}(X, W[1]) = R\Gamma(X, \Omega_X^\bullet)[-1].$$

Algebraic de Rham cohomology of $\mathbb{P}^1$ is $\mathbb{C}$ in degrees zero and two and vanishes otherwise. For completeness, the two-chart Čech complex gives $H^0(\mathcal{O}_X) = \mathbb{C}$, $H^1(\mathcal{O}_X) = 0$, $H^0(\Omega_X^1) = 0$, and $H^1(\Omega_X^1) = \mathbb{C}[dz/z]$. These are the only two entries in its de Rham spectral sequence, so no differential can join them. Normalize $[dz/z]$ by the residue 1 on the overlap. After the displayed shift, these two classes have degrees one and three. Over $\mathbb{C}$, symmetric powers preserve quasi-isomorphisms of complexes, because the symmetric-group coinvariant functors are exact. Together with the first summand of (1.61), their positive symmetric powers give the exterior algebra in (1.60).

Finally, (1.58) identifies the PBW symbol complex with the chiral chains of R . The quasi-isomorphism $\mathcal{B} \rightarrow R$ preserves PBW order and has image in order zero. Applying chiral chains and then (1.52) computes its cohomology by the same exterior algebra. Every positive-order cohomology group therefore vanishes. $\square$

The chiral-algebra zigzag (1.52), the chain map (1.61), and the symbol map (1.58) give the comparison. Every chain arrow has its displayed source and target, including the unit-complex summand. They also specify the next maps after the binary collision: external products on three factors, with compatibility supplied by the chiral Jacobi identity. The two scalar collisions in (1.56) lie below their input PBW orders and are retained in the filtered complex. Passing to the symbol does not assert their vanishing before specialization.

For the fixed-coordinate relative state complex

$$C_{\text{rel}} = \ker b_o \cap \ker H \subset A,$$

the four-state calculation gives $C_{\text{rel}} = \mathbb{C}e \oplus \mathbb{C}v$ with zero differential, in degrees zero and two. It lies in PBW order zero. Consequently it cannot be quasi-isomorphic, with these degrees, to $C^{\text{ch}}(\mathbb{P}^1, \mathcal{A})$ or to its PBW associated graded complex. Their cohomology degrees are those of Theorem 1.29.1. This comparison concerns the unmarked proper curve and algebraic, uncompleted chiral chains. Marked-point coefficients, relative geometric constructions, and inverse-limit completions require their own carriers and comparison maps. The relative Virasoro vacuum complex has classes in degrees zero and two. The full unital Virasoro chiral algebra has chiral-chain classes on the projective line in degrees zero, one, three, and four. A comparison map must therefore lose the relative degree-two class. An explicit one-point chain gives such a map and its null-homotopy.

1.30 THE RELATIVE COMPLEX AND ITS ABSOLUTE PRIMITIVE

Work over $\mathbb{C}$, using algebraic tensor products and direct sums. All complexes are cohomological, with $K[r]^j = K^{j+r}$ and differential $(-1)^r d_K$. Ghost degree is cohomological degree. Conformal weight is a separate integer grading. Interchanging homogeneous factors of degrees a, b contributes $(-1)^{ab}$.

Let $A = \text{Vir}_{26} \otimes bc$ use the universal matter vacuum module and the conformal ghost vacuum Ω . The ghost relations and annihilators are

$$\{b_m, c_n\} = \delta_{m+n,0}, \quad \{b_m, b_n\} = \{c_m, c_n\} = 0, \quad b_n \Omega = 0 \ (n \geq -1), \quad c_n \Omega = 0 \ (n \geq 2).$$

The matter vacuum is implicit in each displayed ghost state. The generators b_n, c_n have degrees $-1, 1$ and weights $-n$. Set

$$Q = \sum_n c_{-n} L_n - \frac{1}{2} \sum_{m,n} (m-n) : c_{-m} c_{-n} b_{m+n} :, \quad H = \mathcal{L}_0 = L_0 + L_0^{\text{gh}}.$$

Normal ordering moves creators to the left and omits contractions. The sums are locally finite on algebraic states. Theorem 1.3.2 and Lemma 1.3.1 give $Q^2 = 0$ and $\{Q, b_o\} = H$.

Fix a coordinate z and define the vector-space complex

$$R = \ker b_o \cap \ker H \subset A.$$

The weight-zero absolute complex has basis

$$e = \Omega, \quad u = c_o \Omega, \quad v = c_{-1} c_1 \Omega, \quad w = c_{-1} c_o c_1 \Omega.$$

Its differential and relative subcomplex are

$$Qe = Qv = Qw = 0, \quad Qu = -2v, \quad (R, Q) = \mathbb{C}e \oplus \mathbb{C}v. \quad (1.62)$$

The degrees of e, u, v, w are 0, 1, 2, 3. Indeed, matter weights are nonnegative, matter weight one vanishes, and the only negative ghost weight is that of c_1 . These facts leave exactly the four displayed weight-zero monomials. The operator b_0 removes the terms containing c_0 . Its anticommutator with Q contracts every nonzero absolute weight. Consequently

$$H^\bullet(R) = \mathbb{C}[e] \oplus \mathbb{C}[v], \quad H^\bullet(A, Q) = \mathbb{C}[e] \oplus \mathbb{C}[w].$$

The formula $v = Q(-u/2)$ determines the relative class image under any comparison that extends to absolute states.

1.31 A NORMALIZED CHAIN IN THE ONE-POINT SUMMAND

Set $X = \mathbb{P}^1$, with ordered cover $U_0 = \text{Spec } \mathbb{C}[z]$, $U_\infty = \text{Spec } \mathbb{C}[z^{-1}]$. Their intersection is $U_{01} = \text{Spec } \mathbb{C}[z, z^{-1}]$. Let $\mathcal{A}^\ell$ be the left state module with connection $\partial_z + \partial$ and let $\mathcal{A} = \omega_X \otimes \mathcal{A}^\ell$. Here ∂ is vertex translation. Coordinate descent and the differential are specified in Section 1.22.

Use the two resolutions P, Q from the construction of (1.55). Explicitly,

$$P^\circ = \text{Sym}_{\mathcal{O}_X}^{\geq 0}(\mathcal{D}_X), \quad P^{-1} = \ker(P^\circ \longrightarrow \mathcal{O}_X), \quad d_P : P^{-1} \hookrightarrow P^\circ.$$

The augmentation on a generating differential operator D is $D(1)$. Write $p = [1_{\mathcal{D}}] \in P^\circ$ for the global generator given by the identity differential operator. Thus $d_P p = 0$ and the augmentation sends p to 1. The symbol p denotes a degree-zero element of the nonunital resolution.

The degree-one term of the two-chart Thom–Sullivan resolution is

$$Q^1 = j_{01*} \mathcal{O}_{U_{01}}[t] dt.$$

In degree zero it consists of triples $(f_0, f_\infty, F(t))$ with $F(0) = f_0$ and $F(1) = f_\infty$ on the overlap. The differential differentiates $F(t)$. Its connection differentiates the coefficients and fixes t, dt . The variable t here is an auxiliary interval coordinate. Normalize $\int_0^1 dt = 1$ and $\text{Res}_{z=0}(dz/z) = 1$.

For a right $\mathcal{D}$ -module N , put $h(N) = N \otimes_{\mathcal{D}_X} \mathcal{O}_X$, degree-wise. The one-point subcomplex of the actual chiral chains is

$$S(\mathcal{A}) = \Gamma(X, h(\mathcal{A} \otimes P \otimes Q))[1] \hookrightarrow \mathbb{C}_X(\mathcal{A}).$$

The target is the Chevalley–Cousin complex (1.55), with its full positive-arity unit carrier. The one-point inclusion is a chain map. A collision merges two distinct blocks, so there is no outgoing collision from a one-block term. This is also immediate from Beilinson–Drinfeld [5], equations (4.2.12.2) and (4.2.12.5), pp. 306–307.

Write ς for suspension, so $|\varsigma x| = |x| - 1$ and $d(\varsigma x) = -\varsigma(dx)$. The section $(dz/z) \otimes p \otimes dt$ is global in the term supported on U_{01} . Define

$$\tau = \varsigma[(dz/z) \otimes p \otimes dt] \in S(\omega_X)^\circ. \quad (1.63)$$

Square brackets in this formula denote the image in h .

LEMMA 1.31.1. The element τ is closed. Its image in $H^0 C_X(\omega_X)$ has trace 1 and is the chiral unit class.

Proof. Both p and dt are closed. The one-point term has no outgoing collision. These observations prove closedness directly in the actual complex.

The right de Rham convention is $\mathrm{DR}_X(\omega_X) = \Omega_X^\bullet[1]$. Consequently $S(\omega_X)$ represents $R\Gamma(X, \Omega_X^\bullet)[2]$. To identify the specific cycle, use the intermediate complex

$$\Gamma(X, \mathrm{DR}_X(\omega_X \otimes P \otimes Q))[1].$$

The same tensor defines a closed element in its terminal Spencer term. The Spencer augmentation maps it to (1.63). The augmentation of P maps it to the form $(dz/z) dt$. The two maps are the resolution comparisons used in Beilinson–Drinfeld, §4.2.12, p. 306. For the right action on ω_X , the Spencer differential sends $f dz \otimes \partial_z$ to $-f' dz$. The identifications $f dz \otimes \partial_z \mapsto f$ and $f dz \mapsto f dz$ give $\mathrm{DR}_X(\omega_X) = \Omega_X^\bullet[1]$. In particular, the top form has its displayed positive sign. Integrating the interval form gives the Čech one-cocycle dz/z with values in Ω_X^1 . It has trace 1 under the stated residue normalization. The trace of the singleton term agrees with the trace on the Cousin diagram by Beilinson–Drinfeld, §4.3.3, pp. 315–316. That proposition identifies $H^0 C_X(\omega_X)$ with $\mathbb{C}$ and all other unit cohomology with zero. Thus the image of τ is the normalized unit class. $\square$

1.32 THE ACTUAL COMPARISON AND ITS CLASS IMAGES

For an algebraic state $a \in A$, let a_z be the constant section in the z -coordinate trivialization over U_0 . Restriction makes it a section over U_{01} . For homogeneous a , define

$$J_z(a) = (-1)^{|a|} \zeta[(dz/z) \otimes a_z \otimes p \otimes dt] \in S(\mathcal{A})^{|a|} \subset C_X(\mathcal{A})^{|a|}. \quad (1.64)$$

Extend by linearity to all algebraic states. All tensors in this formula are finite. The sole pole z^{-1} is regular on the specified overlap. No extension of a_z across infinity is required by Q^1 . For the Poincaré–Birkhoff–Witt (PBW) filtration, give matter and b creators order one and c creators order zero. The auxiliary resolutions have order zero.

PROPOSITION 1.32.1. The map $J_z : (A, Q) \rightarrow C_X(\mathcal{A})$ is a degree-preserving map of vector-space complexes. It preserves the PBW filtration from (1.58). Its restriction $\Phi_z = J_z|_R$ satisfies

$$\Phi_z(e) = 1_{\mathcal{A}*}(\tau), \quad \Phi_z(v) = d(-J_z(u)/2). \quad (1.65)$$

Here $1_{\mathcal{A}*}$ is the actual chain map induced by the chiral unit $\omega_X \rightarrow \mathcal{A}$.

Proof. In the unsuspended tensor, only the differential of a contributes. The remaining factors have $d_P p = 0$ and $d_Q dt = 0$. For $|a| = k$, suspension therefore gives

$$dJ_z(a) = (-1)^{k+1} \zeta[(dz/z) \otimes (Qa)_z \otimes p \otimes dt] = J_z(Qa).$$

Passing to $\hbar$ preserves this equation because Q commutes with the connection. There is no outgoing collision from its one-point image. This proves the chain-map identity in the full complex. The auxiliary factors have PBW order zero, so J_z preserves the state order. The first formula in (1.65) follows by substituting $a = e$. The second follows from $Qu = -2v$ in (1.62). $\square$

The null-homotopy is also explicit as a map of complexes. Let Φ_z° send e to $1_{\mathcal{A}^*}(\tau)$ and v to zero. Define the degree-minus-one linear map $K : R \rightarrow \mathbb{C}_X(\mathcal{A})$ by

$$K(e) = 0, \quad K(v) = -J_z(u)/2.$$

Then $\Phi_z - \Phi_z^\circ = dK + KQ$. These are statements about vector-space complexes and do not assert multiplicativity of J_z .

THEOREM 1.32.2. Use the degree and trace conventions above. Identify the target cohomology by Theorem 1.29.1 as $\Lambda(\xi_1, \xi_3)$, with degrees 1, 3. Then the actual relative comparison (1.64) induces

$$H(\Phi_z)[e] = 1, \quad H(\Phi_z)[v] = 0.$$

Its graded kernel, image, and cokernel are

$$\begin{aligned} \ker H(\Phi_z) &= \mathbb{C}[v], & |[v]| &= 2, \\ \operatorname{im} H(\Phi_z) &= \mathbb{C}1, \\ \operatorname{coker} H(\Phi_z) &= \mathbb{C}\xi_1 \oplus \mathbb{C}\xi_3 \oplus \mathbb{C}\xi_1\xi_3. \end{aligned}$$

The cokernel degrees are 1, 3, 4. The same assertions hold for the induced map to PBW associated graded chiral chains.

For the absolute comparison J_z , normalize ξ_3 by the image of $\tau \otimes \eta$, where $\eta = \alpha\beta\gamma$. Then $H(J_z)[w] = \xi_3$. Thus $H(J_z)$ is injective, with image $\mathbb{C}1 \oplus \mathbb{C}\xi_3$ and cokernel $\mathbb{C}\xi_1 \oplus \mathbb{C}\xi_1\xi_3$.

Proof. The chain identities (1.65) give the relative images. Lemma 1.31.1 and the unital zigzag (1.52) show that the vacuum image is nonzero. The source has precisely the two classes in (1.62). The target basis in Theorem 1.29.1 therefore gives the stated kernel and cokernel.

For the absolute degree-three class, all four weight-zero states lie in the ghost subalgebra B . The one-point construction commutes with the inclusion $B \hookrightarrow A$ and the global evaluation $\epsilon : B \rightarrow E_X$. Here

$$E = \Lambda(\alpha, \beta, \gamma), \quad d\alpha = -\alpha\beta, \quad d\beta = -2\alpha\gamma, \quad d\gamma = -\beta\gamma, \quad f(z) = \alpha + z\beta + z^2\gamma.$$

The identification $S(E_X) = S(\omega_X) \otimes E$ moves E past $P \otimes Q$ with the Koszul sign. The evaluation formulas of (1.51) give

$$\epsilon(e) = 1, \quad \epsilon(u) = -\beta - 2z\gamma, \quad \epsilon(v) = -\alpha\gamma - z\beta\gamma, \quad \epsilon(w) = \eta.$$

In particular, $\epsilon(v) = d((\beta + 2z\gamma)/2)$ at every z . The degree-three image is constant on both charts. After moving the coefficient algebra past dt , the Koszul sign is $(-1)^3$. It cancels the prefactor in (1.64). Hence $\mathbb{C}_X(\epsilon)J_z(w) = \tau \otimes \eta$ in the constant coefficient one-point term.

The generator complex for $\omega_X[-3]$ represents $R\Gamma(X, \Omega_X^\bullet)[-1]$. Its top de Rham class is $\tau \otimes \eta$, of degree three. The positive symmetric-degree comparison (1.61) sends this class to ξ_3 . It is nonzero by that quasi-isomorphism. This determines both absolute class images and their kernel and cokernel.

Finally, e, u, v, w have PBW order zero. The symbol of (1.64) is the same tensor formula with the state replaced by its symbol. The diagram with (1.58) therefore commutes on actual chains. The PBW target cohomology lies entirely in order zero, and its unital comparison has the same normalized classes. This proves the associated graded assertion. $\square$

1.33 COORDINATE DEPENDENCE AND THE RELATIVE DESCENT OBSTRUCTION

The map J_z uses an ordered affine cover, its interval resolution, and the coordinate trivialization on U_0 . The cycle τ is a chosen algebraic representative of the top de Rham class. These choices define a map from the fixed-coordinate state complex to the unmarked global chiral-chain complex. The construction does not require a marked-point coefficient module in the target.

A change of coordinate does not preserve the fixed relative subspace by restriction of the ordinary state transitions. For the convention $z \mapsto z/(1 + sz)$ acting by $g_s = \exp(s\mathcal{L}_1)$,

$$\mathcal{L}_1 v = -c_0 c_1 \Omega, \quad \mathcal{L}_1(c_0 c_1 \Omega) = 0.$$

The ghost commutator $[\mathcal{L}_m, c_n] = (-2m - n)c_{m+n}$ gives both identities. Consequently

$$g_s v = v - s c_0 c_1 \Omega, \quad b_0 g_s v = -s c_1 \Omega, \quad H g_s v = s c_0 c_1 \Omega. \quad (1.66)$$

For $s \neq 0$, this vector is outside R . Thus these transition operators do not define a relative subbundle with model fiber R .

Any map of vector-space complexes from R to this unmarked target must kill its degree-two cohomology class. The target degree-two cohomology vanishes. A map sending $[e]$ to the normalized unit has the class images in Theorem 1.32.2, regardless of its chain representative. Equation (1.65) supplies a primitive for the specific comparison constructed here. Neither this degree obstruction nor (1.66) excludes a different marked or geometric relative complex. Such a complex requires defined coefficient sheaves, transported constraints, and compatible transition maps before a global comparison can be stated.

1.34 A POLYNOMIAL ODD LAPLACIAN

PROPOSITION 1.34.1 (*A polynomial Batalin–Vilkovisky operator*). On $K = \mathbb{C}[x_1, \dots, x_n] \otimes \Lambda(\eta_1, \dots, \eta_n)$, with $|x_i| = 0$ and $|\eta_i| = -1$, use left odd derivatives and put

$$\Delta = \sum_i \partial_{x_i} \partial_{\eta_i}.$$

This operator has degree one, order at most two, and square zero. For homogeneous f, g , define

$$\{f, g\} = (-1)^{|f|} (\Delta(fg) - (\Delta f)g - (-1)^{|f|} f \Delta g).$$

This odd bracket has $\{x_i, \eta_j\} = \delta_{ij}$. For every polynomial $S(x)$, one has $\Delta S = \{S, S\} = 0$, and $D = \{S, -\}$ satisfies $D\Delta + \Delta D = 0$. Thus S solves the polynomial quantum master equation $\frac{1}{2}\{S, S\} + \hbar \Delta S = 0$.

Proof. The even derivatives commute and distinct odd derivatives anticommute; equal odd derivatives square to zero. Pairing indices in Δ^2 proves its vanishing. Each summand is a composition of two derivations and hence has order at most two. Expanding $\Delta(fg)$ gives the degree-one derived bracket; evaluation on generators yields the stated brackets, which determine its biderivation extension. Both ΔS and $\{S, S\}$ vanish since S contains no η . In $D\Delta + \Delta D$, the terms with a derivative left on the input cancel by odd anticommutation, while the remaining Hessian coefficients $\partial_i \partial_j S$ are symmetric and multiply the antisymmetric derivatives $\partial_{\eta_i} \partial_{\eta_j}$. Their sum is zero. $\square$

Remark 1.34.2 (A chiral Batalin–Vilkovisky comparison). The polynomial operator above is defined on its displayed finite generator algebra. A residue operator on a chiral configuration complex requires a domain, a cyclic pairing, its degree and second-order identity, and compatibility with every collision map. The Arnold relations alone do not specify these data or an interacting master action. An identification with the polynomial model requires the actual coefficient and operator maps.

1.35 AN EXPLICIT SUBREGULAR REDUCTION OF $\mathfrak{sl}_3$

The subregular reduction has a nonzero weight-one current and two even charged fields. Their vacuum products determine both the reduction map and its normalization. An integral good grading gives a complex with two charged ghost pairs and no neutral ghosts.

Work first over $R = \mathbb{C}[k]$. All tensor products and direct sums are algebraic. The symbol k is an indeterminate of cohomological degree zero. For matrix units E_{ij} , put

$$a = E_{12}, \quad e = E_{13}, \quad A = E_{21}, \quad B = E_{31}, \quad u = E_{23}, \quad v = E_{32},$$

$$h = \text{diag}(1, -1, 0), \quad j = \frac{1}{3} \text{diag}(1, 1, -2), \quad x = \frac{1}{3} \text{diag}(2, -1, -1), \quad f = -E_{21}.$$

Use the invariant form $(s, t) = \text{tr}(st)$ on $\mathfrak{sl}_3$. The positive eigenspace of $\text{ad } x$ is the abelian algebra $\mathfrak{m} = Ra \oplus Re$, in degree one. The degree-zero space is spanned by h, j, u, v , and the degree-minus-one space by A, B . Direct matrix commutation gives

$$[x, f] = -f, \quad \mathfrak{sl}_3^f = Rj \oplus Ru \oplus RA \oplus RB.$$

Thus every eigenvalue on the centralizer is nonpositive. The character $(f, -)|_{\mathfrak{m}}$ takes values $-1, 0$ on a, e . This is the good-grading convention for quantum reduction in [8, §1, pp. 3–5]. Changing the sign of f does not change its nilpotent orbit.

Let $V_R(\mathfrak{sl}_3)$ be the universal affine vacuum vertex algebra with

$$[s_\lambda t] = [s, t] + k \text{tr}(st) \lambda \mathbf{1}.$$

Tensor it with charged free fermions b_1, b_2, c_1, c_2 , whose only nonzero brackets are

$$[b_i c_j] = [c_j b_i] = \delta_{ij} \mathbf{1}.$$

Affine generators have cohomological degree zero, while $|b_i| = -1$ and $|c_i| = 1$. Parity is cohomological degree modulo two. Write C_R for this differential vertex superalgebra's underlying vertex superalgebra. Use the mode convention

$$Y(s, z) = \sum_{n \in \mathbb{Z}} s_{(n)} z^{-n-1}, \quad [s_\lambda t] = \sum_{n \geq 0} \frac{\lambda^n}{n!} s_{(n)} t, \quad : st := s_{(-1)} t.$$

All generating modes with $n \geq 0$ annihilate the vacuum. Every state is a finite sum of ordered negative-mode monomials.

Define the odd field and its zero mode by

$$q =: (a - \mathbf{1})c_1 : + : ec_2 :, \quad D = q_{(0)}.$$

The residue of $z^{-1}dz$ is one. There are no singular contractions in $q(z)q(w)$, so $D^2 = 0$. The zero-mode commutator formula makes D a derivation of every vertex product and gives $D\partial = \partial D$. In particular,

$$\begin{aligned} Db_1 &= a - \mathbf{1}, & Db_2 &= e, & Dc_i &= Da = De = 0, \\ Dh &= -2 : ac_1 : - : ec_2 :, & Dj &= - : ec_2 :, & Du &= : ec_1 :, \\ Dv &= : ac_2 :, & DA &= : hc_1 : - : uc_2 : + k\partial c_1, \\ DB &= - : vc_1 : + \frac{1}{2} : (h + 3j)c_2 : + k\partial c_2. \end{aligned}$$

1.35.1 *Four closed fields*

Set $N_i = b_i c_i$: and define

$$H = h + 2N_1 + N_2, \quad J = j + N_2, \quad U = u - : b_2 c_1 :, \quad X = v - : b_1 c_2 :. \quad (1.67)$$

Let C_R^- be the vertex subalgebra generated by $H, J, U, X, A, B, c_1, c_2$. Its cohomological degrees are nonnegative.

LEMMA 1.35.1. The subalgebra C_R^- is preserved by D . Its differential satisfies

$$\begin{aligned} DH &= -2c_1, & DX &= c_2, & DJ &= DU = Dc_i = 0, \\ DA &= : Hc_1 : - : Uc_2 : + (k+3)\partial c_1, \\ DB &= - : Xc_1 : + \frac{1}{2} : (H + 3J)c_2 : + (k+3)\partial c_2. \end{aligned} \quad (1.68)$$

The following four even states are closed:

$$\begin{aligned} J, \quad U, \quad P &= A + \frac{1}{4} : HH : + : UX : + \frac{k+2}{2} \partial H, \\ V &= B - \frac{1}{2} : HX : - \frac{3}{2} : JX : - (k+2)\partial X. \end{aligned} \quad (1.69)$$

Proof. The affine brackets and the ghost brackets show that C_R^- is a vertex subalgebra with a PBW basis in its displayed generators. For example,

$$[H_\lambda H] = (2k+5)\lambda \mathbf{1}, \quad [H_\lambda J] = \lambda \mathbf{1}, \quad [U_\lambda X] = -\frac{1}{2}H + \frac{3}{2}J + (k+1)\lambda \mathbf{1}.$$

The bracket of A or B with a modified current remains a linear combination of A, B . The brackets of modified degree-zero currents close among themselves and scalar terms. Their brackets with c_i are linear combinations of the c_i .

Normal ordering gives the identities

$$\begin{aligned} : N_i c_i : &:= -\partial c_i, & : N_2 c_1 : + : (: b_2 c_1 :) c_2 : &:= -\partial c_1, \\ : (: b_1 c_2 :) c_1 : + : N_1 c_2 : &:= -\partial c_2. \end{aligned}$$

They follow by moving the rightmost ghost mode through the two modes of the normal product. Substitution in DA, DB proves (1.68). The quasi-commutativity formula gives

$$: c_1 H : - : H c_1 : = 2\partial c_1, \quad : c_1 X : - : X c_1 : = -\partial c_2.$$

Therefore

$$\begin{aligned} D(\frac{1}{4} : HH :) &= - : H c_1 : - \partial c_1, & D(: UX :) &= : U c_2 :, \\ D(-\frac{1}{2} : HX :) &= : c_1 X : - \frac{1}{2} : H c_2 :, & D(-\frac{3}{2} : JX :) &= -\frac{3}{2} : J c_2 :. \end{aligned}$$

The remaining derivative terms in DP, DV cancel exactly. $\square$

Let W_R denote the vertex subalgebra of C_R^- generated by J, U, P, V , with zero differential.

THEOREM I.35.2 (*Subregular reduction map*). The inclusion

$$\iota : (W_R, \mathfrak{o}) \hookrightarrow (C_R, D)$$

is a strict unital quasi-isomorphism of differential vertex superalgebras over R . Ordered normally ordered monomials in all derivatives of J, U, P, V form an R -basis of W_R . For every $k_o \in \mathbb{C}$, specialization gives

$$H^r(C_R \otimes_R \mathbb{C}_{k_o}, D) = \begin{cases} W_R \otimes_R \mathbb{C}_{k_o}, & r = \mathfrak{o}, \\ \mathfrak{o}, & r \neq \mathfrak{o}. \end{cases}$$

Here $\mathbb{C}_{k_o} = R/(k - k_o)$, including $k_o = -3$. The degree-zero vertex algebra is the subregular quantum reduction for $(\mathfrak{sl}_3, x, f)$.

Proof. Let C_R^+ be generated by $a - \mathfrak{1}, e, b_1, b_2$. Its vertex products are commutative and its differential is preserved. Normally ordered multiplication gives a linear map of complexes

$$C_R^- \otimes_R C_R^+ \longrightarrow C_R, \quad s \otimes t \longmapsto s_{(-1)}t.$$

The tensor differential is $D(s \otimes t) = Ds \otimes t + (-1)^{|s|} s \otimes Dt$. To prove bijectivity, assign filtration degrees zero to derivatives of a, e, b_i , one to derivatives of h, j, u, v, c_i , and two to derivatives of A, B . All singular brackets lower total filtration degree by at least one. The associated graded is the polynomial algebra in the even jets tensored with the exterior algebra in the odd jets. The changes $a \mapsto a - \mathfrak{1}$ and (1.67), including all derivatives, are polynomial and invertible there. Thus multiplication induces a bijection on associated graded modules. Induction on the nonnegative filtration degree proves the stated linear bijection. The filtration need not have finite-dimensional pieces. Every individual state has finite degree. This is a tensor decomposition of complexes, not of vertex algebras.

Write $a'_r = \partial^r(a - \mathfrak{1})$, $e_r = \partial^r e$, and $b_{ir} = \partial^r b_i$. The complex C_R^+ has differential

$$D_+ = \sum_{r \geq \mathfrak{o}} \left(a'_r \frac{\partial}{\partial b_{1r}} + e_r \frac{\partial}{\partial b_{2r}} \right).$$

Odd derivatives act from the left. For $s_+ = \sum_r (b_{1r} \partial_{a'_r} + b_{2r} \partial_{e_r})$, one has $D_+ s_+ + s_+ D_+ = N_+$, where N_+ counts these factors. Division by each positive integer eigenvalue gives a contraction onto $R\mathfrak{1}$. All sums are finite on a state. Consequently $C_R^- \hookrightarrow C_R$ is a quasi-isomorphism.

On C_R^- assign degree one to derivatives of H, J, U, X, c_i , and degree two to derivatives of A, B . The associated graded differential is

$$dH = -2c_1, \quad dX = c_2, \quad dJ = dU = dc_i = \mathfrak{o},$$

$$dA = Hc_1 - Uc_2, \quad dB = -Xc_1 + \frac{1}{2}(H + 3J)c_2.$$

It commutes with all formal derivatives. Introduce the polynomial variables

$$p = A + H^2/4 + UX, \quad v_o = B - HX/2 - 3JX/2.$$

This triangular change is invertible, and $dp = dv_o = \mathfrak{o}$. Hence the graded complex is

$$R[\partial^r J, \partial^r U, \partial^r p, \partial^r v_o : r \geq \mathfrak{o}] \otimes_R (R[H_r, X_r] \otimes \Lambda(c_{1r}, c_{2r}), d).$$

On the second factor,

$$s_- = -\frac{1}{2} \sum_r H_r \frac{\partial}{\partial c_{1r}} + \sum_r X_r \frac{\partial}{\partial c_{2r}}$$

satisfies $ds_- + s_- d = N_-$. Thus the graded cohomology is precisely the first polynomial factor in degree zero.

The explicit cycles (1.69) lift its four generators. Their ordered monomials have independent leading symbols, which proves linear independence. Given a degree-zero closed state, subtract a linear combination of these monomials with the same leading symbol. The filtration degree decreases. Repeating terminates and proves that these monomials span $\ker(D|_{(C_R^-)^\circ})$. There are no degree-minus-one states in C_R^- . Thus this kernel is W_R itself. For a closed state in positive cohomological degree, the graded contraction produces a preimage of its leading symbol. Subtracting its differential lowers the filtration degree and again terminates. It follows that $H^{r>0}(C_R^-) = 0$.

These arguments use only division by nonzero integers and the constant 2. They therefore apply over R and after every specialization. No inverse of $k+3$ enters this proof. This proves all assertions, including base change and the identification with the defining quantum reduction. $\square$

Theorem 1.35.2 gives an explicit instance of the structure theorem in [8, Theorem 4.1 and its proof, pp. 10–12]. The displayed contraction proves the required assertions on this universal vacuum carrier directly. Write $C_k = C_R \otimes_R \mathbb{C}_k$ and $W_k = W_R \otimes_R \mathbb{C}_k$ for a fixed complex level.

1.35.2 The Bershadsky–Polyakov products

For $k \neq -3$, set $t = k+3$ and

$$\begin{aligned} G^+ &= U, & G^- &= -V, & T &= \frac{P + \frac{3}{4} : JJ : - \frac{3}{2} \partial J}{t}, \\ \kappa_J &= \frac{2k+3}{3}, & c &= -\frac{(2k+3)(3k+1)}{k+3}. \end{aligned} \tag{1.70}$$

The sign in $G^- = -V$ fixes the cubic vacuum term. The fields $G^\pm$ are even.

PROPOSITION 1.35.3. The following brackets hold as identities of states in C_k , not only in cohomology:

$$\begin{aligned} [J_\lambda J] &= \kappa_J \lambda \mathbf{1}, & [J_\lambda G^\pm] &= \pm G^\pm, & [G^\pm_\lambda G^\pm] &= 0, \\ [T_\lambda J] &= (\partial + \lambda)J, & [T_\lambda G^\pm] &= (\partial + \frac{3}{2}\lambda)G^\pm, \\ [T_\lambda T] &= (\partial + 2\lambda)T + \frac{c}{12} \lambda^3 \mathbf{1}. \end{aligned} \tag{1.71}$$

$$\begin{aligned} [G^+_\lambda G^-] &= 3 : JJ : + \frac{3(k+1)}{2} \partial J - tT + 3(k+1)\lambda J \\ &\quad + \frac{(k+1)(2k+3)}{2} \lambda^2 \mathbf{1}. \end{aligned} \tag{1.72}$$

Consequently W_k is the universal Bershadsky–Polyakov vertex algebra in this normalization.

Proof. Use the noncommutative Wick formula

$$[s_\lambda : ab :] = [s_\lambda a]b + (-1)^{p(s)p(a)} : a[s_\lambda b] : + \int_0^\lambda [[s_\lambda a]_\mu b] d\mu$$

and quasi-commutativity : $ab : -(-1)^{p(a)p(b)} : ba := \int_{-\partial}^{\circ} [a_{\lambda} b] d\lambda$. For $M_{ij} =: b_i c_j$;, their input ghost identities are

$$\begin{aligned} [M_{ij\lambda} M_{lm}] &= \delta_{jl} M_{im} - \delta_{im} M_{lj} + \delta_{jl} \delta_{im} \lambda \mathbf{1}, \\ [M_{ij\lambda} b_l] &= \delta_{jl} b_i, \quad [M_{ij\lambda} c_l] = -\delta_{il} c_j. \end{aligned}$$

Affine-ghost mixed brackets vanish. These formulas immediately give $[J_{\lambda} J] = \kappa_J \lambda$ and $[J_{\lambda} U] = U$. The next substitutions give

$$\begin{aligned} [J_{\lambda} V] &= -V, \\ [J_{\lambda} P] &= \frac{3}{2} \lambda J + \frac{2k+3}{2} \lambda^2 \mathbf{1}, \\ [U_{\lambda} V] &= P - \frac{2}{4} : JJ : - \frac{3(k+2)}{2} \partial J - 3(k+1) \lambda J - \frac{(k+1)(2k+3)}{2} \lambda^2 \mathbf{1}. \end{aligned}$$

For the last identity, insert $V = B - : HX : / 2 - 3 : JX : / 2 - (k+2) \partial X$. The required brackets are

$$[U_{\lambda} B] = A, \quad [U_{\lambda} H] = U, \quad [U_{\lambda} J] = -U, \quad [U_{\lambda} X] = -H/2 + 3J/2 + (k+1) \lambda.$$

The two Wick integrals combine to $\lambda U_{(0)} X + \frac{k+1}{2} \lambda^2 \mathbf{1}$. Expanding the four displayed summands, and using quasi-commutativity for H, J , yields the asserted formula. Substitution of (1.70) proves (1.72). The formula for $[J_{\lambda} P]$ and $[J_{\lambda} : JJ :] = 2\kappa_J \lambda J$ give the stated T, J bracket.

For the conformal normalization, let S be the affine Sugawara field and put

$$L^{\text{full}} = S + \partial x + \sum_{i=1}^2 : (\partial b_i) c_i : .$$

Its affine generator weights are $1 - \deg_x(s)$. Each ghost pair has weights $(0, 1)$ and central charge -2 . The field q is primary of weight one, so $DL^{\text{full}} = 0$. The Sugawara and ghost contractions give

$$c_L = \frac{8k}{t} - 8k - 4, \quad [L^{\text{full}}_{\lambda} J] = (\partial + \lambda) J - \frac{\kappa_J}{2} \lambda^2 \mathbf{1}.$$

Here $(x, x) = 2/3$, $(x, j) = 1/3$, and the second ghost pair contributes $-\lambda^2/2$. There is an explicit homotopy identity:

$$t(L^{\text{full}} - T - \frac{1}{2} \partial J) = D(: b_1 A : + : b_2 B :). \quad (1.73)$$

For its verification, write the affine Sugawara numerator as

$$tS = \frac{1}{4} : hh : + \frac{3}{4} : jj : + \frac{1}{2} (: aA : + : Aa : + : eB : + : Be : + : uv : + : vu :).$$

Use $: Aa : =: aA : - \partial h$, $: Be : =: eB : - \partial(h + 3j)/2$, and $: vu : =: uv : + \partial(h - 3j)/2$. Substitute (1.67)–(1.69). The remaining terms are $: (a - 1)A : + : eB : - : b_1 DA : - : b_2 DB :$, which equal the right side of (1.73) by the odd derivation rule.

Improving L^{full} by $-\partial J/2$ changes the central charge by $3\kappa_J$. It makes J primary and gives $G^{\pm}$ weights $3/2$. Here charge is the eigenvalue of $J_{(0)}$. Under $L^{\text{full}}_{(1)}$, the states H, J, U, X have weight one and A, B, P, V have weight two. The improvement adds half the charge to these weights. Its cohomology class is $[T]$. The inclusion $\mathcal{W}_k \rightarrow C_k$ is injective on cohomology. Thus its conformal identities hold strictly in $\mathcal{W}_k$. The value $c = c_L + 3\kappa_J$ is exactly (1.70). Higher products $T_{(n)} G^{\pm}$, $n \geq 2$, have charge ± 1 and weight at most $1/2$. The PBW basis has no such states. Products $G^{\pm}_{(n)} G^{\pm}$, $n \geq 0$, have charge ± 2 and weight at most two. The smallest PBW weight with either charge is three. These products therefore vanish. This proves all brackets. They agree with the universal presentation in [9, §2, p. 2]. The PBW theorem proves that this presentation introduces no further relations. $\square$

COROLLARY 1.35.4 (*Special levels and reflection*). The current vacuum coefficient is $\kappa_J = (2k + 3)/3$. The G^+ , G^- cubic coefficient is $(k + 1)(2k + 3)$, and the T , T quartic coefficient is $c/2$. All three vanish simultaneously at $k = -3/2$. At $k = -1/2$ they are $2/3, 1, 1/5$, respectively. For $k' = -k - 6$ and $k \neq -3$, the reflected conformal charges obey

$$c(k) + c(k') = 50.$$

Equality $c(k) = c(k')$ holds precisely when $(k + 3)^2 = -4$. Hence a conformal vertex isomorphism at reflected levels is impossible for all other noncritical parameters.

Proof. The common root follows from the first coefficient and direct substitution in the other two. At $k = -1/2$, the charge is $2/5$. Writing $t = k + 3$ gives $c(k) = 25 - 6t - 24/t$ and $c(k') = 25 + 6t + 24/t$. Their equality is equivalent to $t^2 = -4$. A conformal unit map preserves $T_{(3)}T = (c/2)\mathbf{1}$. At the two exceptional parameters, this calculation states only equality of charges. It does not construct an isomorphism. $\square$

1.35.3 Unital binary collisions

Fix a complex level k , including $k = -3$, and an affine coordinate z on $\mathbb{A}^1$. The map $\iota : W_k \rightarrow C_k$ commutes with translation and every vertex product. For a differential vertex algebra F , its right differential-operator module is $\mathcal{A}_F = \omega_{\mathbb{A}^1} \otimes_{\mathbb{C}} F$, with

$$(g(z) dz \otimes a) \cdot \partial_z = -g'(z) dz \otimes a - g(z) dz \otimes \partial a.$$

In left-module notation, put

$$\mathcal{P}_F = \mathbb{C}[z, w, (z - w)^{-1}] \otimes F \otimes F, \quad \mathcal{R}_F = F \otimes \mathbb{C}[w] \otimes \xi^{-1} \mathbb{C}[\xi^{-1}], \quad \xi = z - w.$$

The source connections are $\partial_z^{\text{base}} + \partial_F \otimes \mathbf{1}$ and $\partial_w^{\text{base}} + \mathbf{1} \otimes \partial_F$. The target actions are

$$\partial_z = \partial_\xi, \quad \partial_w = \partial_w^{\text{base}} - \partial_\xi + \partial_F.$$

Multiplication by ξ on the target is taken modulo regular powers. Tensoring with $dz \wedge dw$ gives the corresponding right modules. The orientation is $dz \wedge dw = d\xi \wedge dw$, with $\text{Res } \xi^{-1} d\xi = \mathbf{1}$. The tensor differential on $\mathcal{P}_F$ uses the sign $(-1)^{|a|}$ on its second state. Define the full meromorphic collision by

$$\mu_F(g(z, w) \otimes a \otimes b) = \text{pp}_{\xi=0}(g(w + \xi, w)Y(a, \xi)b).$$

Here pp retains the strictly negative powers of ξ . The answer is finite by vertex truncation and the finite pole order of g . Both μ_F and ι commute with the internal differential.

PROPOSITION 1.35.5. The map $\mathcal{A}_{W_k} \rightarrow \mathcal{A}_{C_k}$, $g dz \otimes a \mapsto g dz \otimes \iota(a)$, is a unital quasi-isomorphism of differential chiral algebras on this affine chart. Its tensorwise maps make the square of μ_{W_k} and μ_{C_k} commute exactly. They are quasi-isomorphisms on $\mathcal{P}$ and $\mathcal{R}$, and on their collision mapping cones. In particular, the vacuum terms are preserved:

$$\mu_{C_k}(J \otimes J) = \kappa_J \xi^{-2} \mathbf{1},$$

and, for $k \neq -3$,

$$\begin{aligned} \mu_{C_k}(G^+ \otimes G^-) &= (k + 1)(2k + 3) \xi^{-3} \mathbf{1} + 3(k + 1) \xi^{-2} J \\ &\quad + \xi^{-1} \left(3 : JJ : + \frac{3(k+1)}{2} \partial J - (k + 3)T \right). \end{aligned}$$

Proof. Equality of every mode product gives the commutative square coefficient by coefficient. The identities $Y(\partial a, \xi) = \partial_\xi Y(a, \xi)$ and $[\partial_F, Y(a, \xi)] = \partial_\xi Y(a, \xi)$ prove compatibility with both connections. The vertex skew-symmetry and Jacobi identities give the chiral identities after taking principal parts and converting to right modules. The vacuum embedding is $g dz \mapsto g dz \otimes \mathbf{1}$. Its collision is $\text{pp}_\xi(\xi^{-1}Y(\mathbf{1}, \xi)a) = \xi^{-1}a$. Localization and tensor products over $\mathbb{C}$ preserve quasi-isomorphisms. The same holds for the direct sum $\xi^{-1}\mathbb{C}[\xi^{-1}]$. For a cochain map $\mu : P \rightarrow R$, use $\text{Cone}(\mu)^n = R^n \oplus P^{n+1}$ and $d(r, p) = (d_R r + \mu p, -d_P p)$. The cone's short exact sequence gives the cone quasi-isomorphism. The two displayed collisions follow directly from (1.71)–(1.72). The same coefficient argument preserves every finite iterated vertex product. $\square$

There is no strict unital vertex map $C_k \rightarrow \mathcal{W}_k$ preserving cohomological degree. It would kill b_i, c_i and contradict $b_{i(0)}c_i = \mathbf{1}$. The inverse to ι is therefore an inverse in the category obtained by inverting quasi-isomorphisms. The polynomial family and its binary collision maps specify no comparison with the reflected Koszul dual. A completed chiral bar, a second differential, and descent through nodal curves require their own constructions.

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